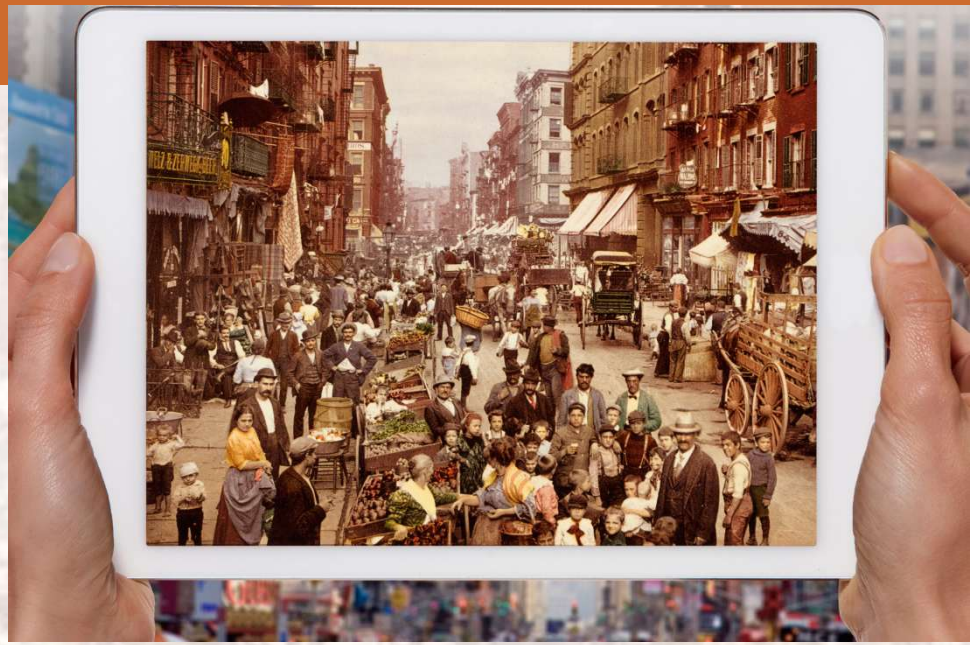
經濟學



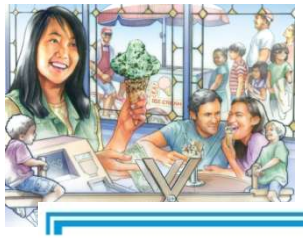
CHAPTER 4

The Market Forces of Supply and Demand

National Central University

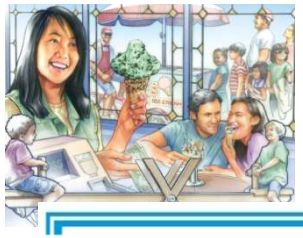
Look for the answers to these questions

- What factors affect buyers' demand for goods?
- What factors affect sellers' supply of goods?
- How do supply and demand determine the price of a good and the quantity sold?
- How do changes in the factors that affect demand or supply affect the market price and quantity of a good?
- How do markets allocate resources?



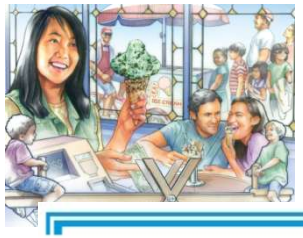
Markets and Competition

- Market
  - A group of buyers and sellers of a particular good or service
  - Buyers as a group
    - Determine the demand for the product
  - Sellers as a group
    - Determine the supply of the product



Markets and Competition

- Competitive market
  - Many buyers and many sellers, each has a negligible impact on market price
- Perfectly competitive market （完全競争市場）
  - All goods are exactly the same
  - Buyers and sellers are so numerous that no one can affect the market price, “Price takers”



Demand

- Quantity demanded
  - Amount of a good that buyers are willing and able to purchase
- Law of demand
  - Other things equal 其他條件不變的情況下
  - When the price of a good rises, the quantity demanded of the good falls
  - When the price falls, the quantity demanded rises

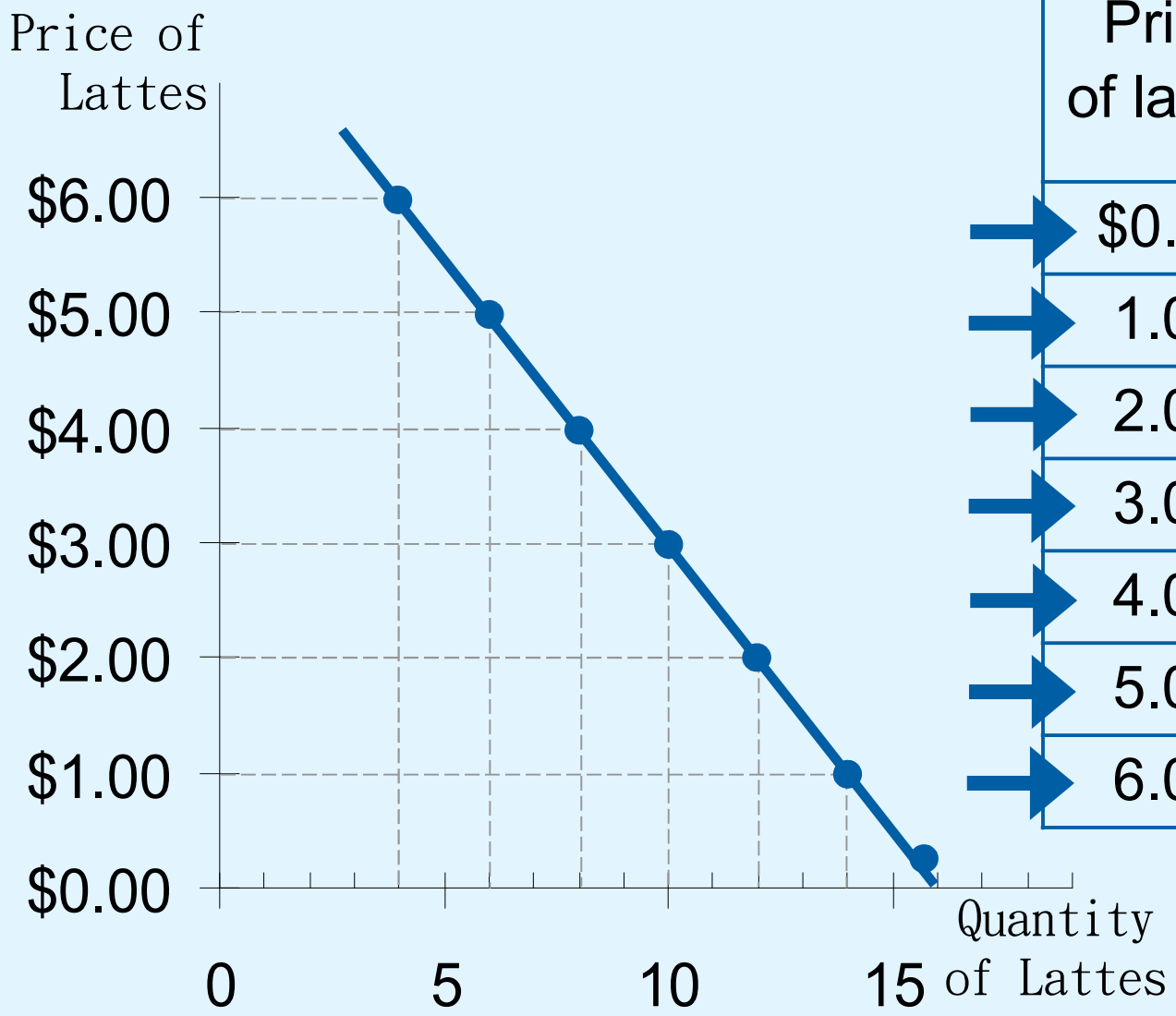
Sam's Demand Schedule

Demand schedule:

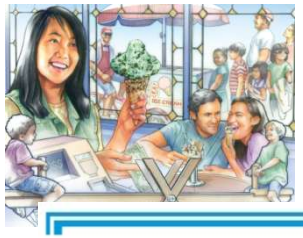
- A table, shows the relationship between the price of a good and the quantity demanded
- Example: Sam's demand for lattes
- Notice that Sam's preferences obey the law of demand.

Price of lattes	Quantity of lattes demanded
$0.00	16
1.00	14
2.00	12
3.00	10
4.00	8
5.00	6
6.00	4

Sam's Demand Schedule and Demand Curve



Price of lattes	Quantity of lattes demanded
$0.00	16
1.00	14
2.00	12
3.00	10
4.00	8
5.00	6
6.00	4



Demand

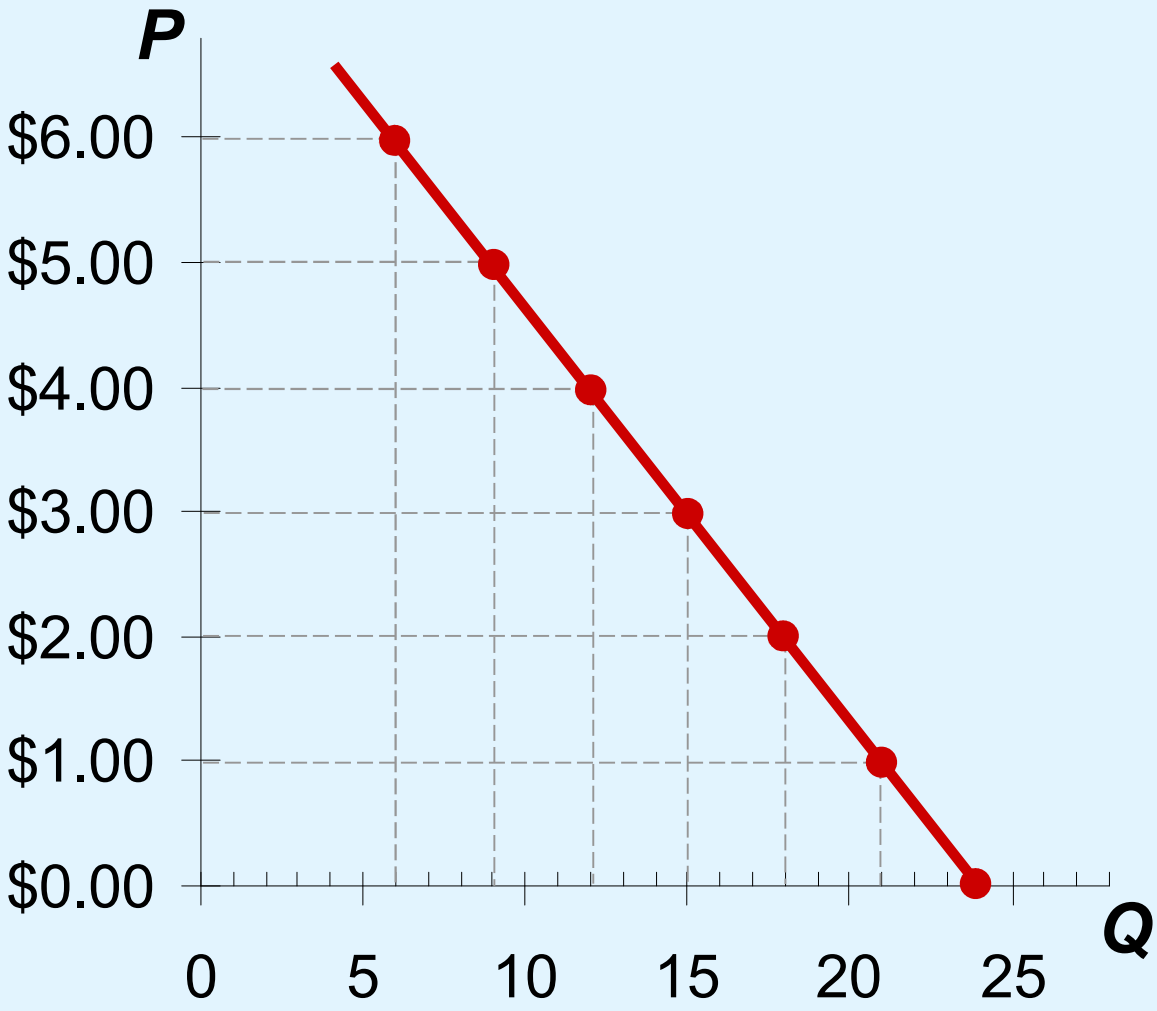
- Market demand
  - Sum of all individual demands for a good or service
  - Market demand curve: sum the individual demand curves horizontally
    - To find the total quantity demanded at any price, we add the individual quantities

Market Demand versus Individual Demand

Suppose Sam and Dean are the only two buyers in the market for lattes. (Q^d = quantity demanded)

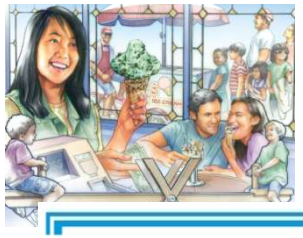
Price	Sam's Q^d		Dean's Q^d		Market Q^d
$0.00	16	+	8	=	24
1.00	14	+	7	=	21
2.00	12	+	6	=	18
3.00	10	+	5	=	15
4.00	8	+	4	=	12
5.00	6	+	3	=	9
6.00	4	+	2	=	6

The Market Demand Curve for Lattes



P	Q^d (Market)
$0.00	24
1.00	21
2.00	18
3.00	15
4.00	12
5.00	9
6.00	6

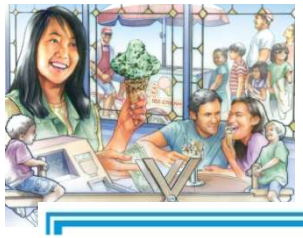
Q^d 為 Q^{demand} 的縮寫



Demand Curve Shifters

- The demand curve
  - Shows how price affects quantity demanded, other things being equal
- These "other things" are non-price determinants of demand
  - Things that determine buyers' demand for a good, other than the good's price
- Changes in them shift the D curve…

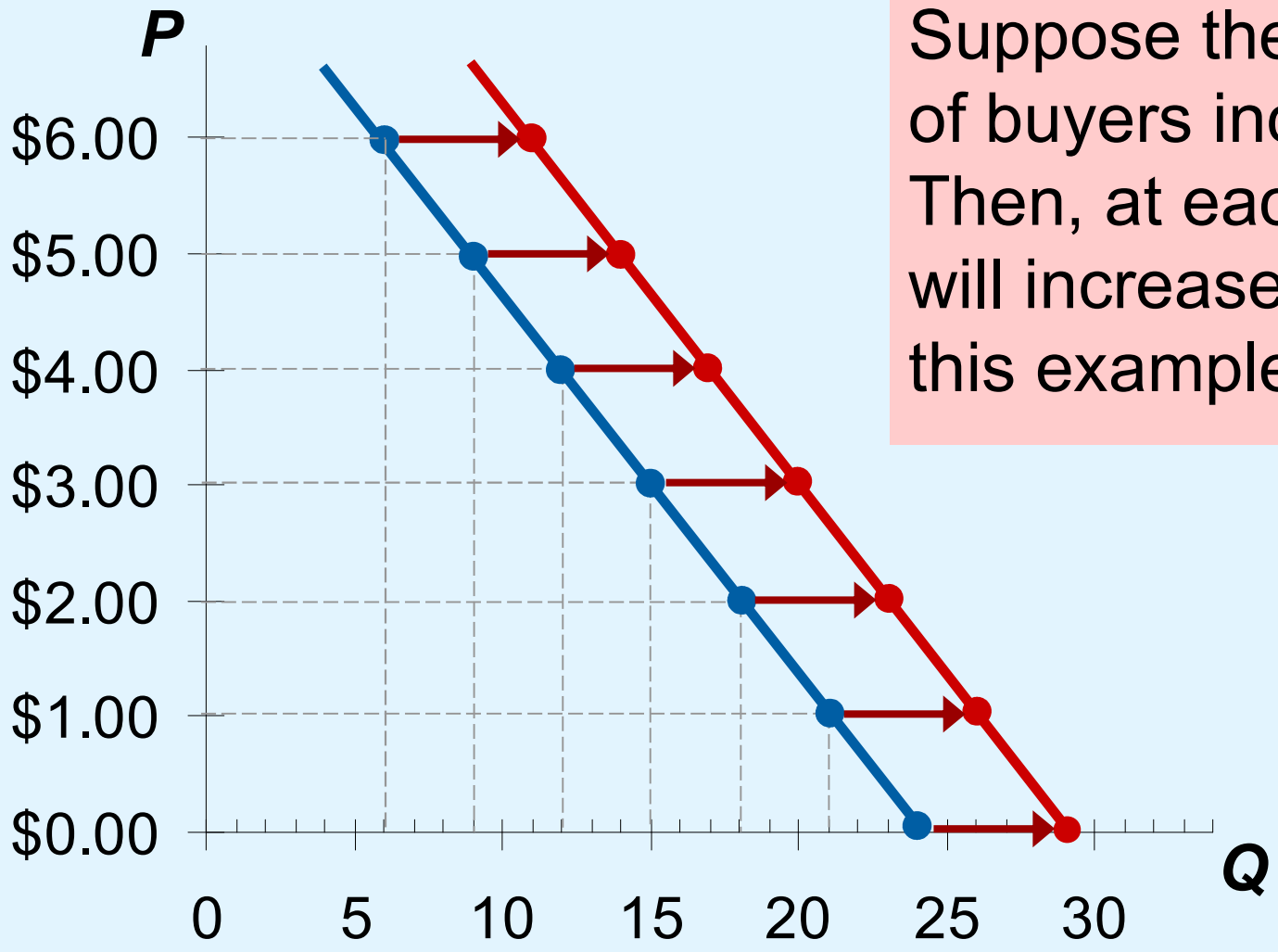
非價格因素



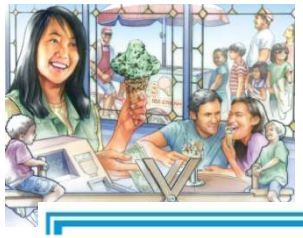
Demand Curve Shifters

- Number of buyers
  - Increase in # of buyers
    - Increases quantity demanded at each price
    - Shifts D curve to the right
  - Decrease in # of buyers
    - Decreases quantity demanded at each price
    - Shifts D curve to the left

Demand Curve Shifters: # of Buyers



Suppose the number of buyers increases. Then, at each P, Q^d will increase (by 5 in this example).



Demand Curve Shifters

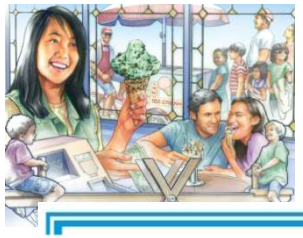
- Income
  - Normal good, other things constant
    - An increase in income leads to an increase in demand: Shifts D curve to the right
  - Inferior good, other things constant
    - An increase in income leads to a decrease in demand: Shifts D curve to the left

劣質商品範例罕見, 地瓜可能可以視為劣質商品
早年台灣收入低落, 以地瓜為主食(地瓜買多)
現在台灣經濟發達, 以米飯為主食(地瓜少買)



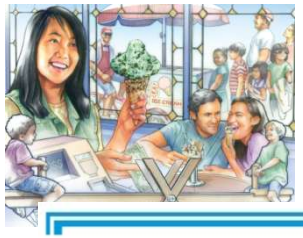
Substitutes





Substitutes





Complements

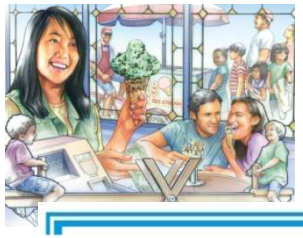




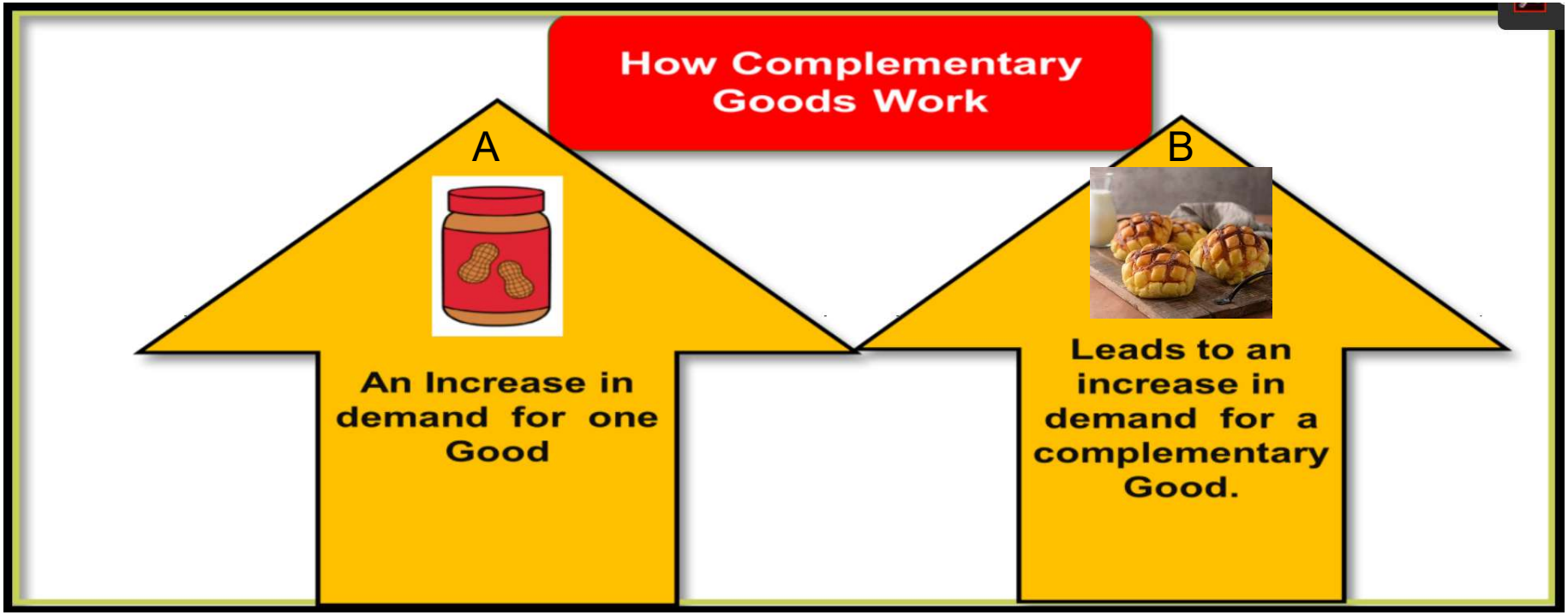


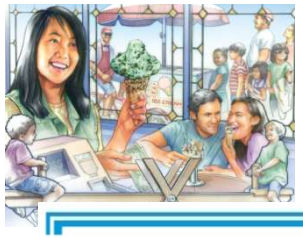






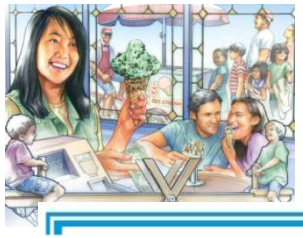
Complements





Demand Curve Shifters

- Prices of related goods, substitutes
  - Two goods are substitutes if
    - An increase in the price of one leads to an increase in the demand for the other
  - Example: pizza and hamburgers
    - An increase in the price of pizza increases demand for hamburgers, *shifting hamburger demand curve to the right*
  - Other examples:
    - Coke and Pepsi, laptops and tablets, music CDs and music downloads

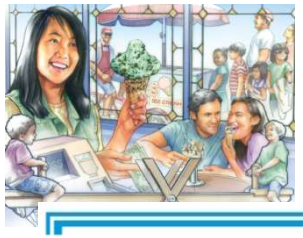


Demand Curve Shifters

- Prices of related goods, complements
  - Two goods are complements if
    - An increase in the price of one leads to a decrease in the demand for the other
  - Example: computers and software
    - If price of computers rises, people buy fewer computers, and therefore less software; *Software demand curve shifts left*
  - Other examples:
    - College tuition and textbooks, bagels and cream cheese, eggs and bacon

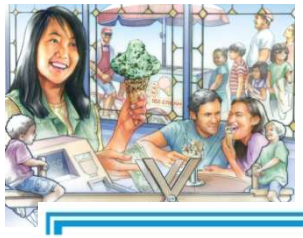
阿金飲食法


全民愛健康
飲食篇12
認識阿金飲食減重法
圖文創作：健談 havemary.com
專家諮詢：財團法人彰化基督教醫院 陳燕華營養師
第一期 誘導期
第二期 持續減重期
第三期 維持前期
第四期 維持期
主要攝取肉、蛋、油脂與綠色蔬菜，限制醣類攝取加速脂肪分解，約持續兩週
兩星期約可減重10%
醣類每星期增加2克，找出減重臨界最高醣量
醣類每星期繼續增加，找出不減重也不增重的臨界醣量
找出適合自己的飲食內容；若體重增加，可返回前期來減重
一般而言，二、三期的醣類應在每天50 克左右
健談 havemary.com
健康九九網站 關心您的健康



Demand Curve Shifters

- Tastes
  - Anything that causes a shift in tastes toward a good will increase demand for that good and shift its D curve to the right
  - Example:
    - The Atkins diet (阿金飲食法) became popular in the ’90s, caused an increase in demand for eggs, shifted the egg demand curve to the right



Demand Curve Shifters

- Expectations about the future
  - Expect an increase in income, increase in current demand
  - Example:
    - If people expect their incomes to rise, their demand for meals at expensive restaurants may increase now
  - Expect higher prices increase in current demand
    - e.g., CPC announces oil price hike next week

Upcoming Tariff Reduction



Summary: Variables That Influence Buyers

Variable	A Change in This Variable . . .
Price of the good itself	Represents a movement along the demand curve
Income	Shifts the demand curve
Prices of related goods	Shifts the demand curve
Tastes	Shifts the demand curve
Expectations	Shifts the demand curve
Number of buyers	Shifts the demand curve

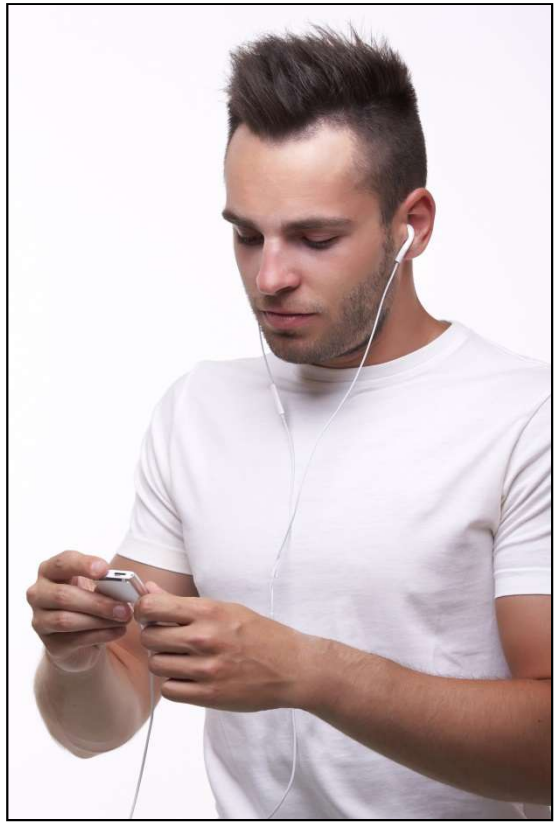
Active Learning 1 — Demand curve

- Draw a demand curve for music downloads
- What happens to it in each of the following scenarios?
- Why?

A. The price of iPods falls

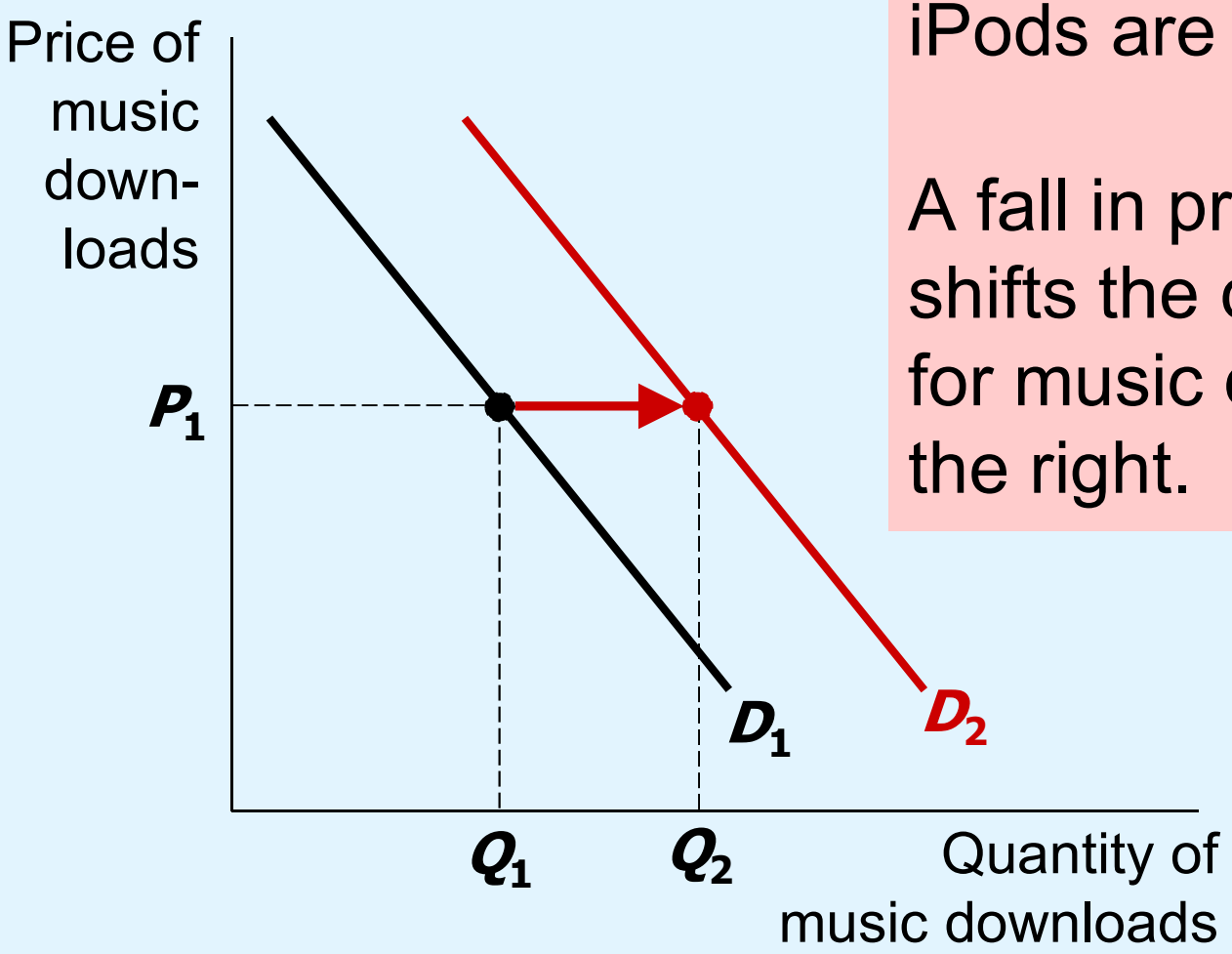
B. The price of music downloads falls

C. The price of music CDs falls



Active Learning 1

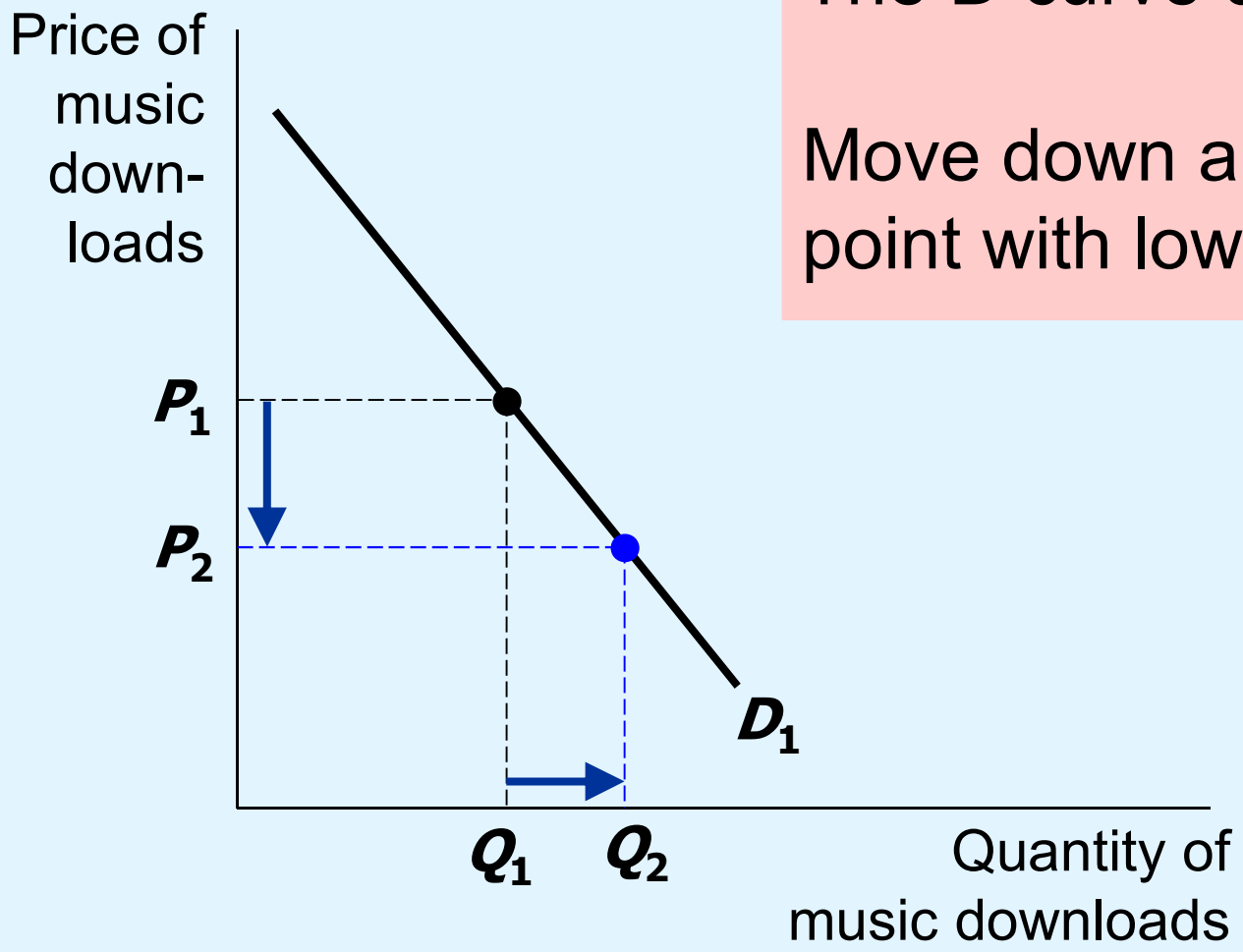
A. The price of iPods falls



Music downloads and iPods are complements.

A fall in price of iPods shifts the demand curve for music downloads to the right.

Active Learning 1 **B.** The price of music downloads falls

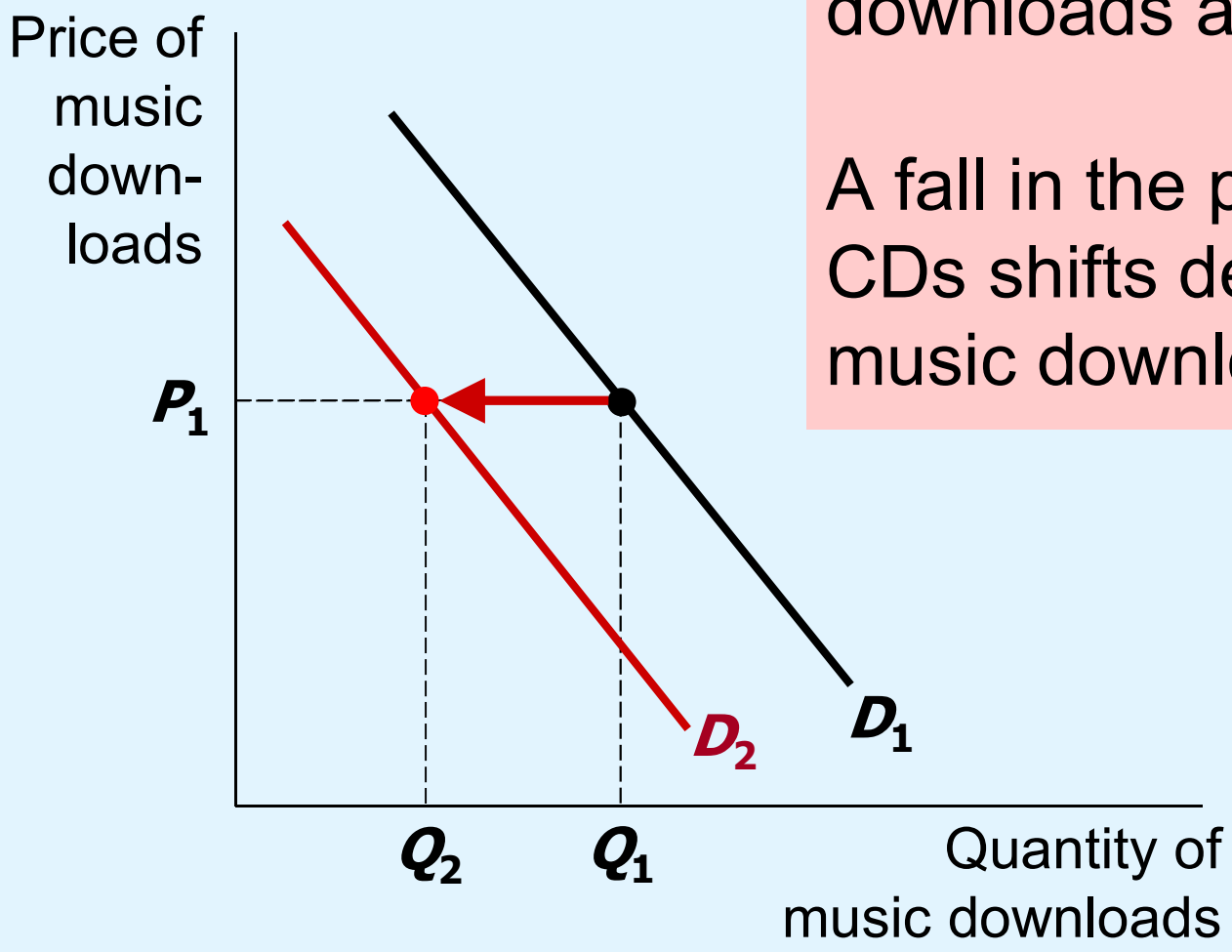


The D curve does not shift.

Move down along curve to a point with lower P, higher Q.

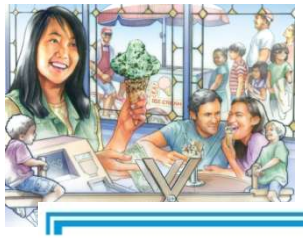
Active Learning 1

C. The price of music CDs falls



Music CDs and music downloads are substitutes.

A fall in the price of music CDs shifts demand for music downloads to the left.



Supply

- Quantity supplied
  - Amount of a good
  - Sellers are willing and able to sell
- Law of supply
  - Other things equal
  - When the price of a good rises, the quantity supplied of the good rises
  - When the price falls, the quantity supplied falls

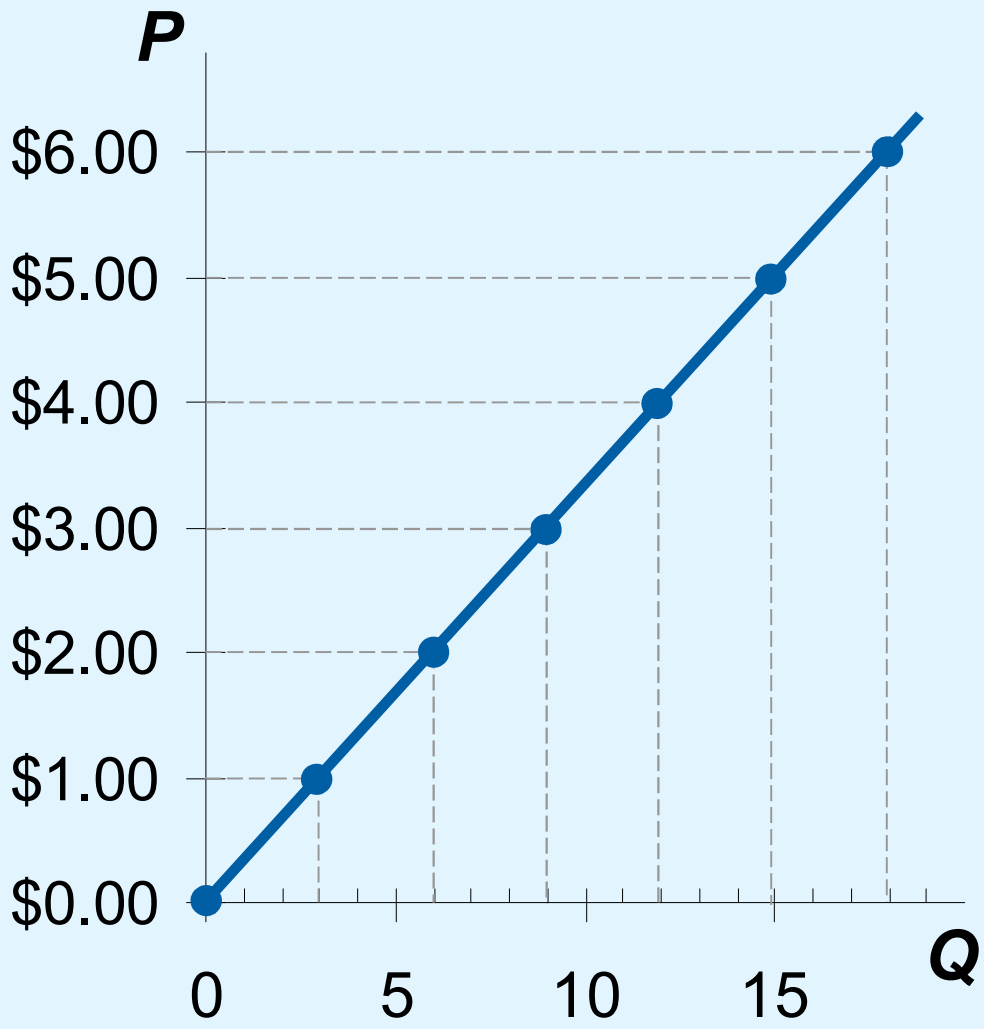
Starbucks' Supply Schedule

Supply schedule:

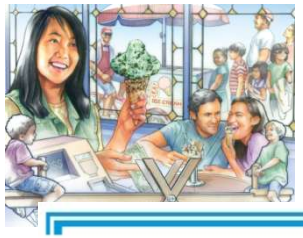
- A table, shows the relationship between the price of a good and the quantity supplied.
- Example: Starbucks' supply of lattes
- Notice that Starbucks' supply schedule obeys the law of supply

Price of lattes	Quantity of lattes supplied
$0.00	0
1.00	3
2.00	6
3.00	9
4.00	12
5.00	15
6.00	18

Starbucks' Supply Schedule and Supply Curve



Price of lattes	Quantity of lattes supplied
$0.00	0
1.00	3
2.00	6
3.00	9
4.00	12
5.00	15
6.00	18



Market Supply vs. Individual Supply

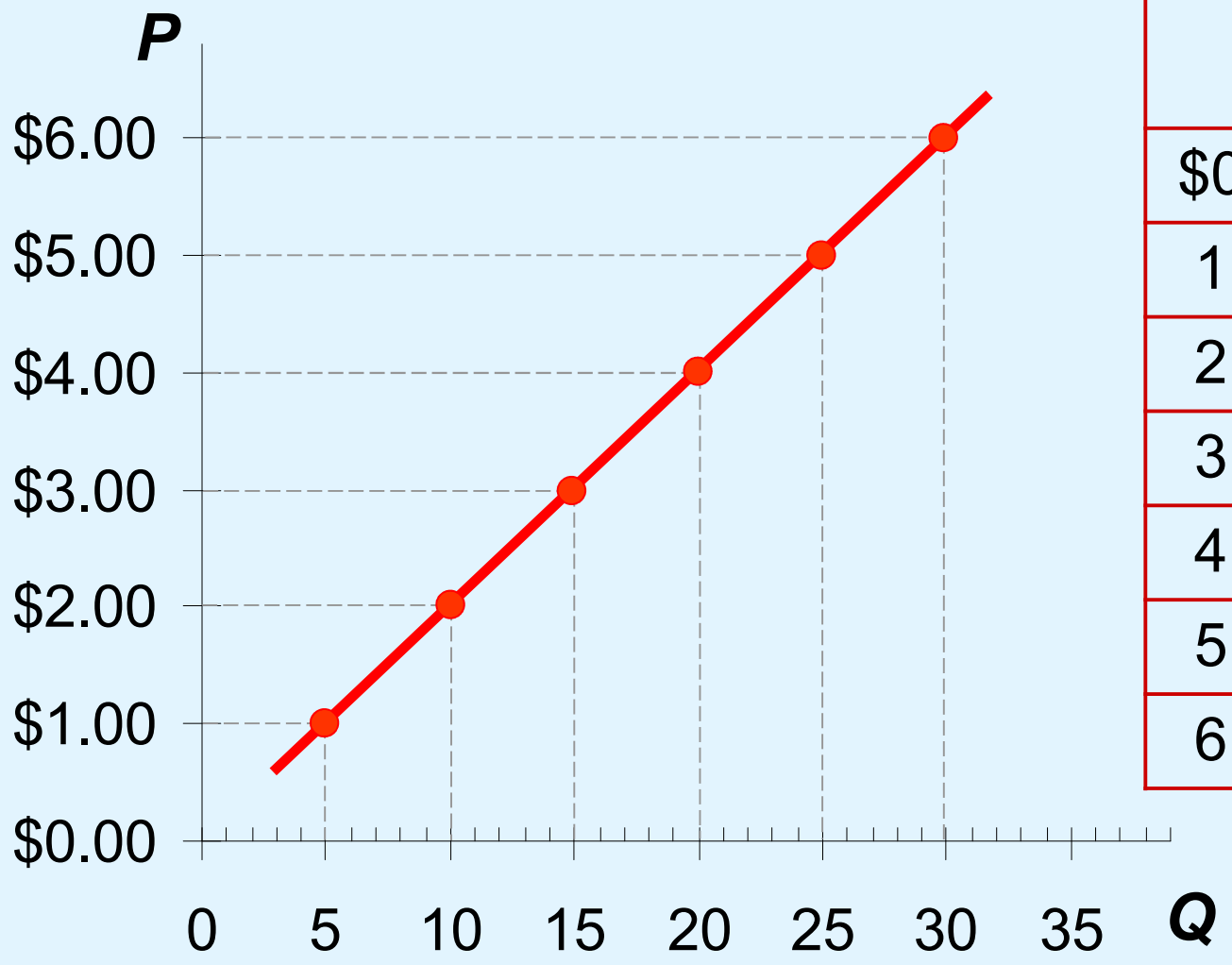
- Market supply
  - Sum of the supplies of all sellers of a good or service
  - Market supply curve: sum of individual supply curves horizontally
    - To find the total quantity supplied at any price, we add the individual quantities

Market Supply vs. Individual Supply

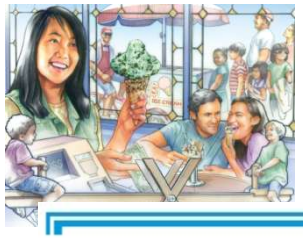
Suppose Starbucks and Peet's are the only two sellers in this market. (Q^s = quantity supplied)

Price	Q^s Starbucks		Q^s Peet's		Market Q^s
$0.00	0	+	0	=	0
1.00	3	+	2	=	5
2.00	6	+	4	=	10
3.00	9	+	6	=	15
4.00	12	+	8	=	20
5.00	15	+	10	=	25
6.00	18	+	12	=	30

The Market Supply Curve

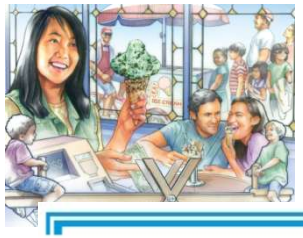


P	Q^S (Market)
$0.00	0
1.00	5
2.00	10
3.00	15
4.00	20
5.00	25
6.00	30



Supply Curve Shifters

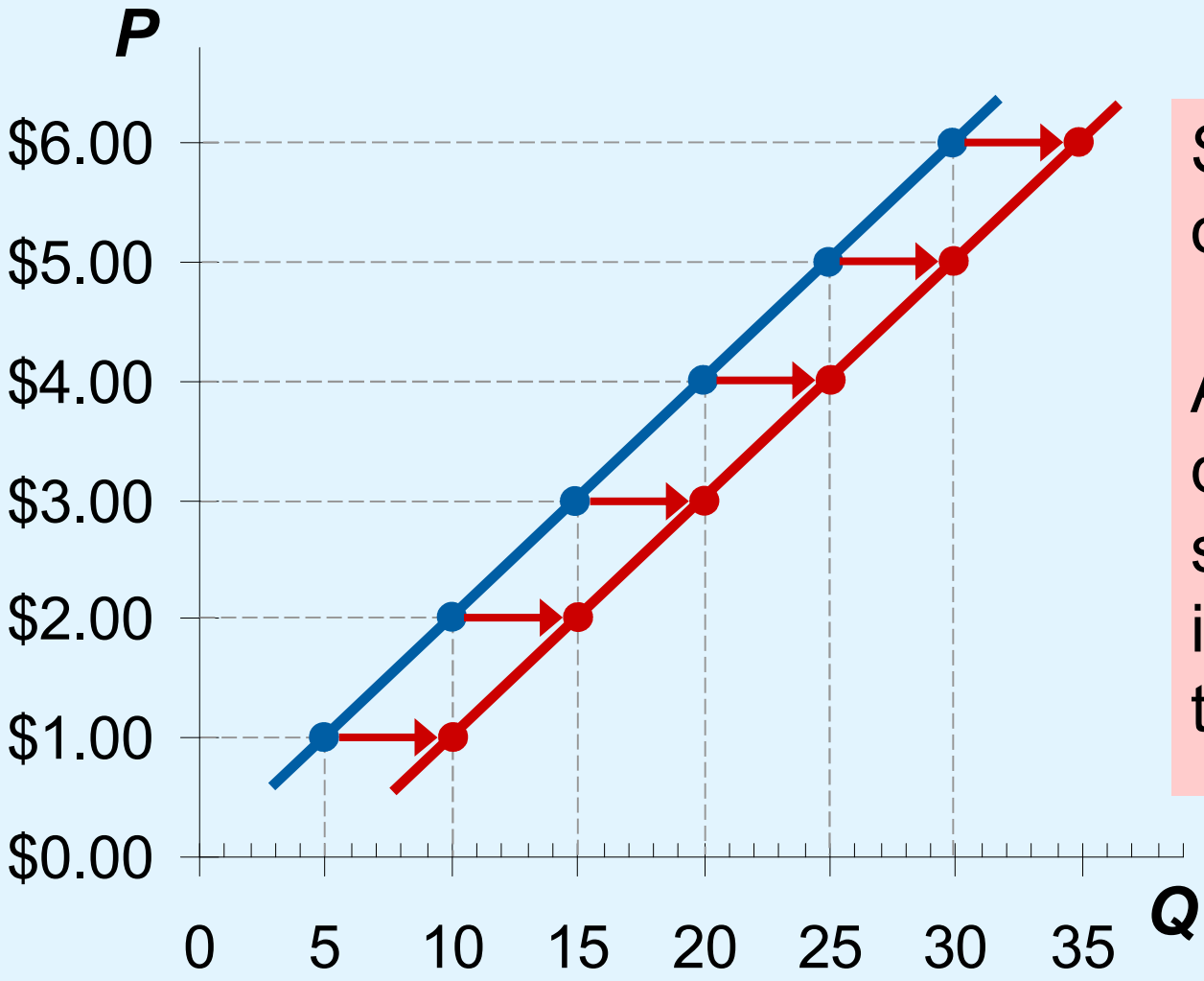
- The supply curve
  - Shows how price affects quantity supplied, other things being equal
- These "other things"
  - Are non-price determinants of supply
- Changes in them shift the S curve…



Supply Curve Shifters

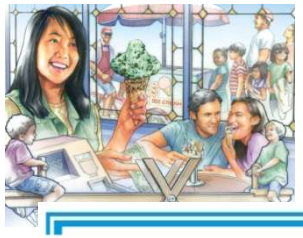
- Input prices
  - Supply is negatively related to prices of inputs
  - Examples of input prices: wages, prices of raw materials
  - A fall in input prices makes production more profitable at each output price
    - Firms supply a larger quantity at each price
    - The S curve shifts to the right

Supply Curve Shifters: Input Prices



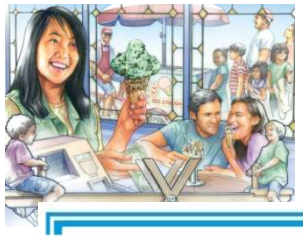
Suppose the price of milk falls.

At each price, the quantity of lattes supplied will increase (by 5 in this example).



Supply Curve Shifters

- Technology
  - Determines how much inputs are required to produce a unit of output
    - e.g., energy-saving device
  - A cost-saving technological improvement has the same effect as *a fall in input prices*, shifts S curve to the right
- Number of sellers
  - An increase in the number of sellers
    - Increases the quantity supplied at each price
    - Shifts S curve to the right



Supply Curve Shifters

- Expectations about future
  - Example: Events in the Middle East lead to expectations of higher oil prices
    - Owners of Texas oil fields reduce supply now, save some inventory to sell later at the higher price
    - S curve shifts left
  - Sellers may adjust supply* when their expectations of future prices change (*If good not perishable) 非生鮮食品

Summary: Variables That Influence Sellers

Variable	A Change in This Variable . . .
Price of the good itself	Represents a movement along the supply curve
Input prices	Shifts the supply curve
Technology	Shifts the supply curve
Expectations	Shifts the supply curve
Number of sellers	Shifts the supply curve

U.S. Tax Refund Policy

Form **1040** Department of the Treasury—Internal Revenue Service (99) **U.S. Individual Income Tax Return** **2016** OMB No. 1545-0074 IRS Use Only—Do not write or staple in this space.

For the year Jan. 1–Dec. 31, 2016, or other tax year beginning , 2016, ending , 20 See separate instructions.

Your first name and initial	Last name	Your social security number
If a joint return, spouse's first name and initial	Last name	Spouse's social security number
Home address (number and street). If you have a P.O. box, see instructions.	Apt. no.	▲ Make sure the SSN(s) above and on line 6c are correct.
City, town or post office, state, and ZIP code. If you have a foreign address, also complete spaces below (see instructions).		Presidential Election Campaign
Foreign country name	Foreign province/state/county / Foreign postal code	Check here if you, or your spouse if filing jointly, want $3 to go to this fund. Checking a box below will not change your tax or refund. ☐ You ☐ Spouse

Filing Status Check only one

1 ☐ Single
2 ☐ Married filing jointly (even if only one had income)
3 ☐ Married filing separately. Enter spouse's SSN above
4 ☐ Head of household (with qualifying person). (See instructions.) If the qualifying person is a child but not your dependent, enter this child's name here. ▶



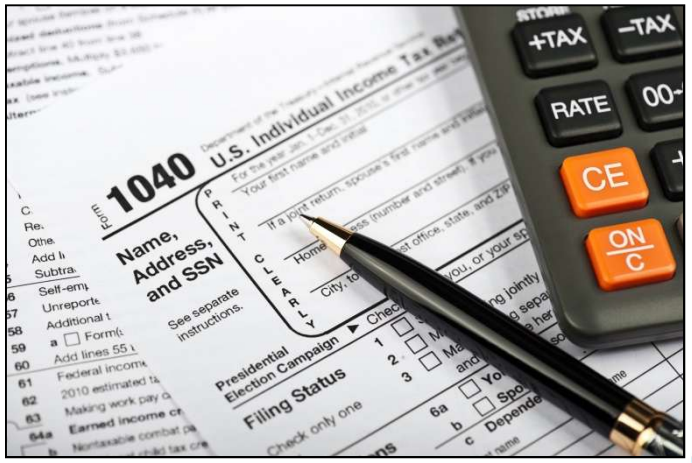
Active Learning 2 — Supply curve

Draw a supply curve for tax return preparation software (退稅試算軟體). What happens to it in each of the following scenarios?

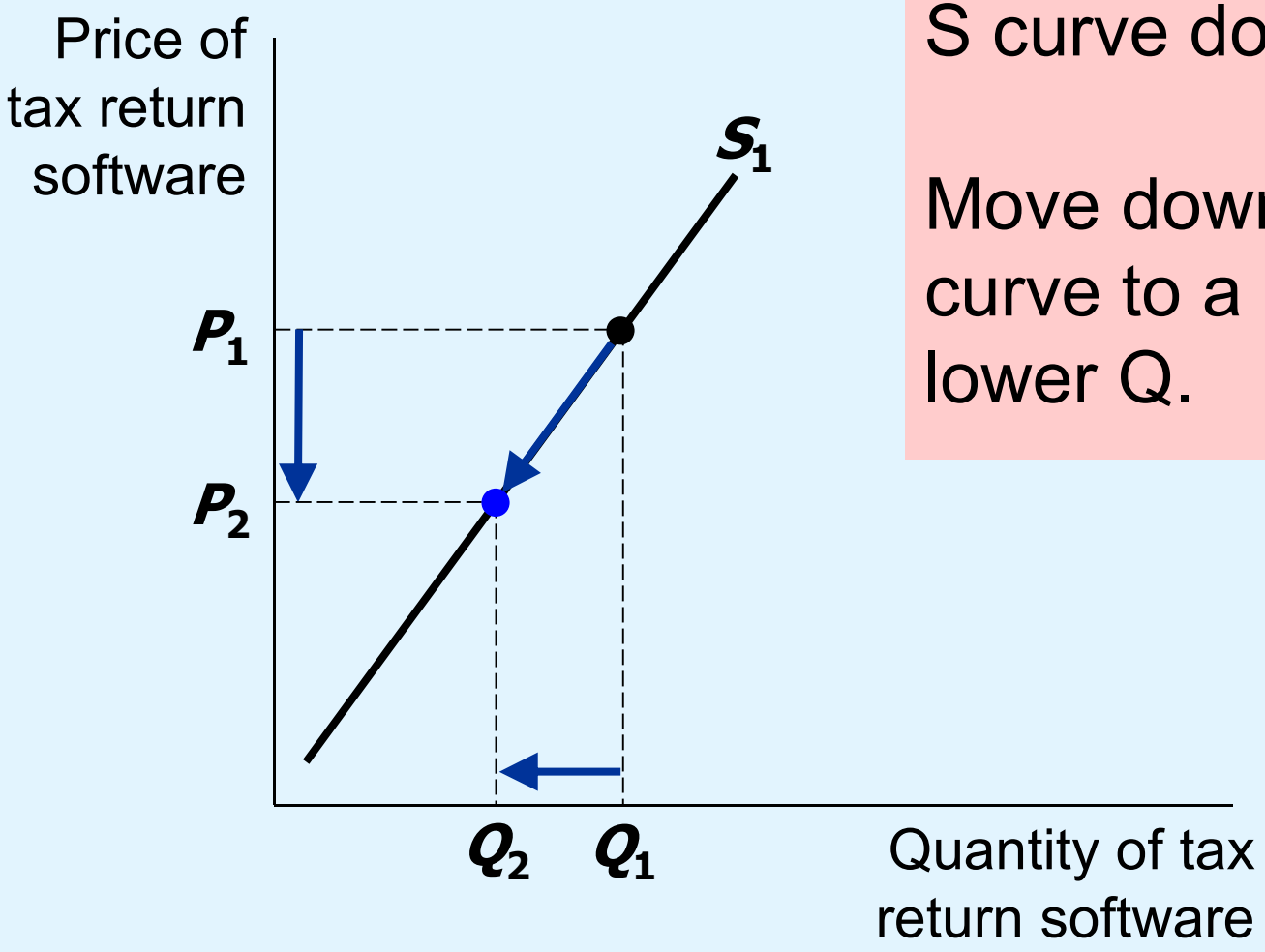
A. Retailers cut the price of the software.

B. A technological advance allows the software to be produced at lower cost.

C. Professional tax return preparers raise the price of the services they provide.



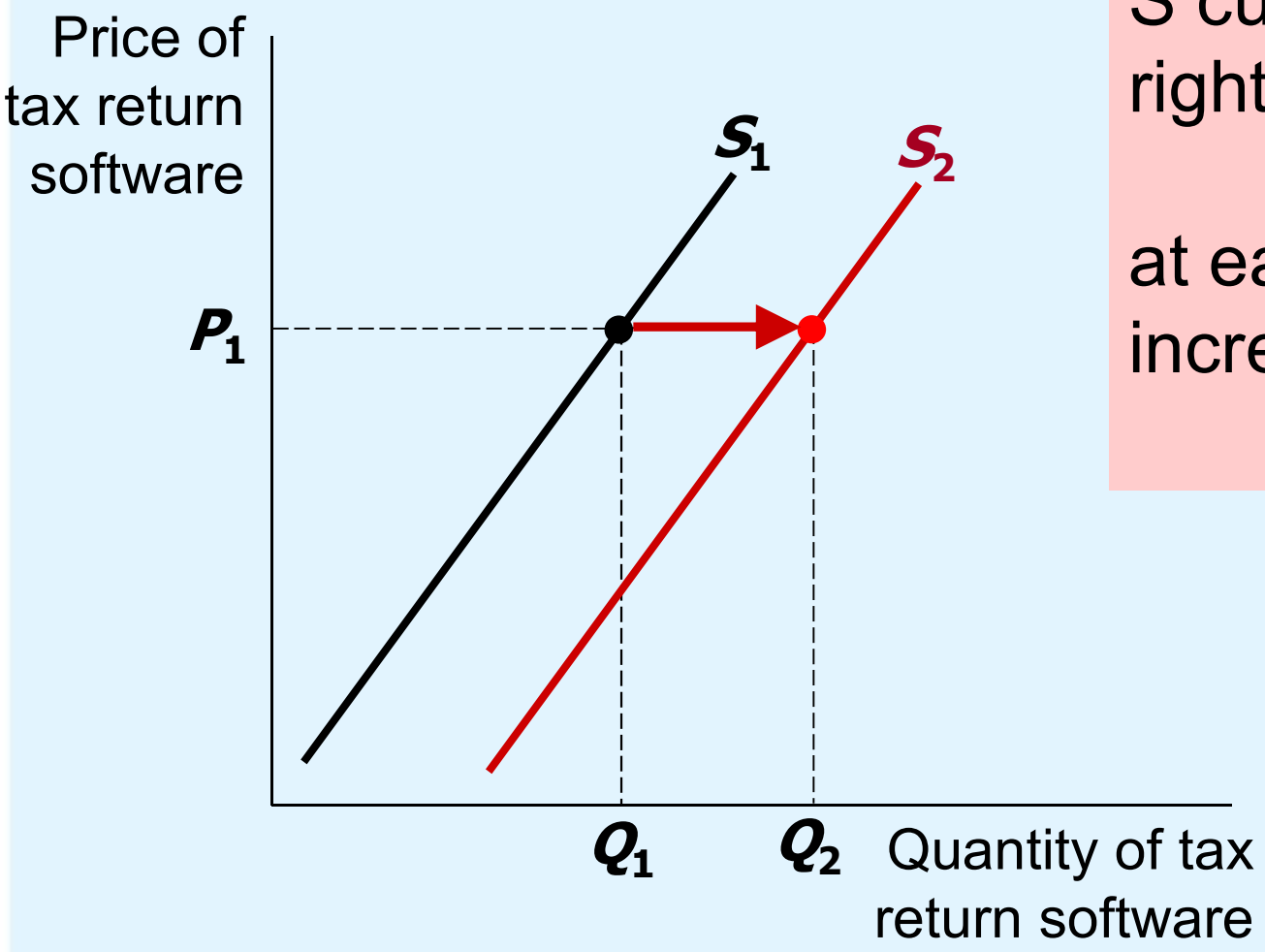
Active Learning 2 **A.** Fall in price of tax return software



S curve does not shift.

Move down along the curve to a lower P and lower Q.

Active Learning 2 **B.** Fall in cost of producing software

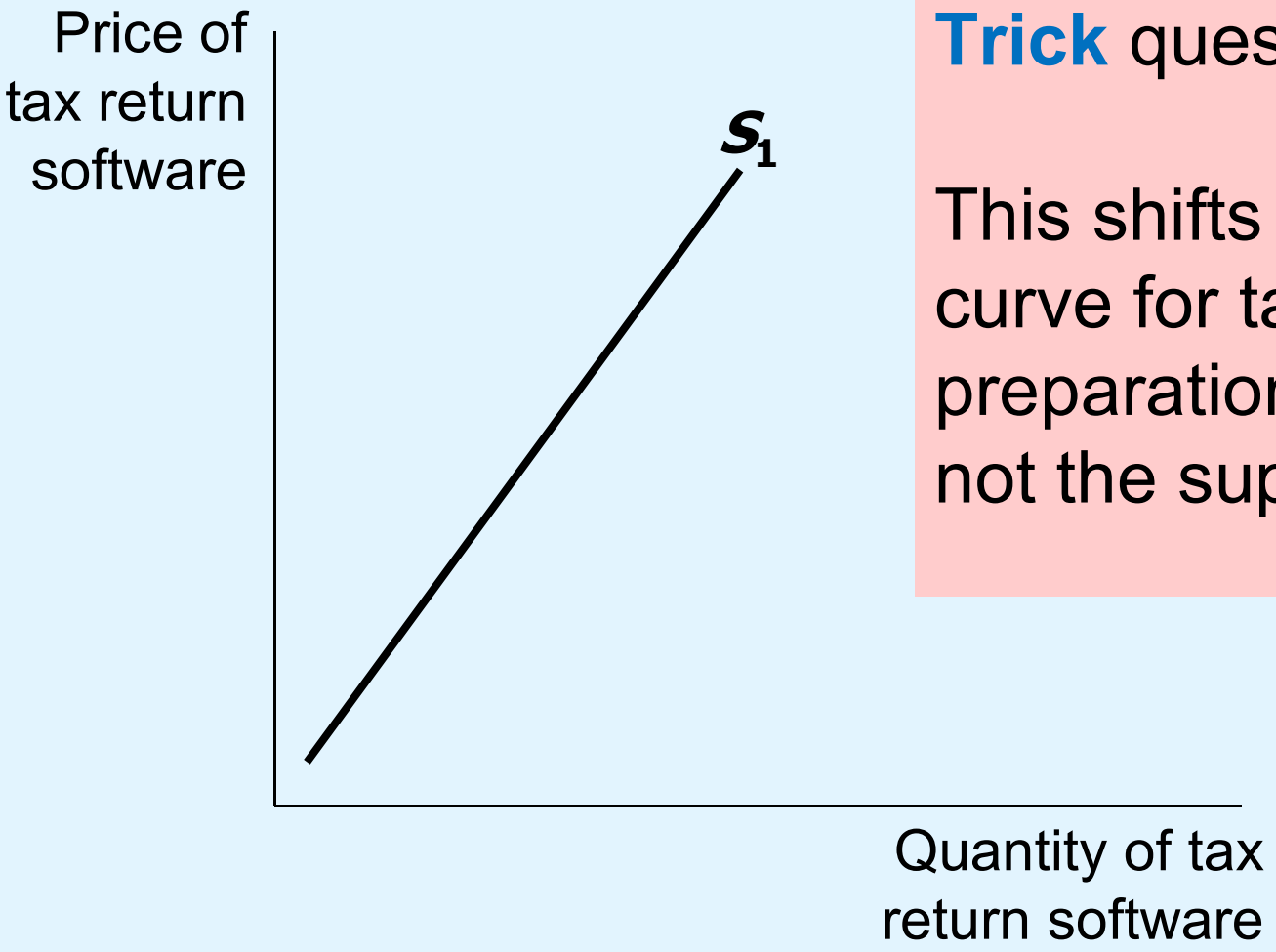


S curve shifts to the right:

at each price, Q increases.

Active Learning 2

C. Professional preparers raise their price

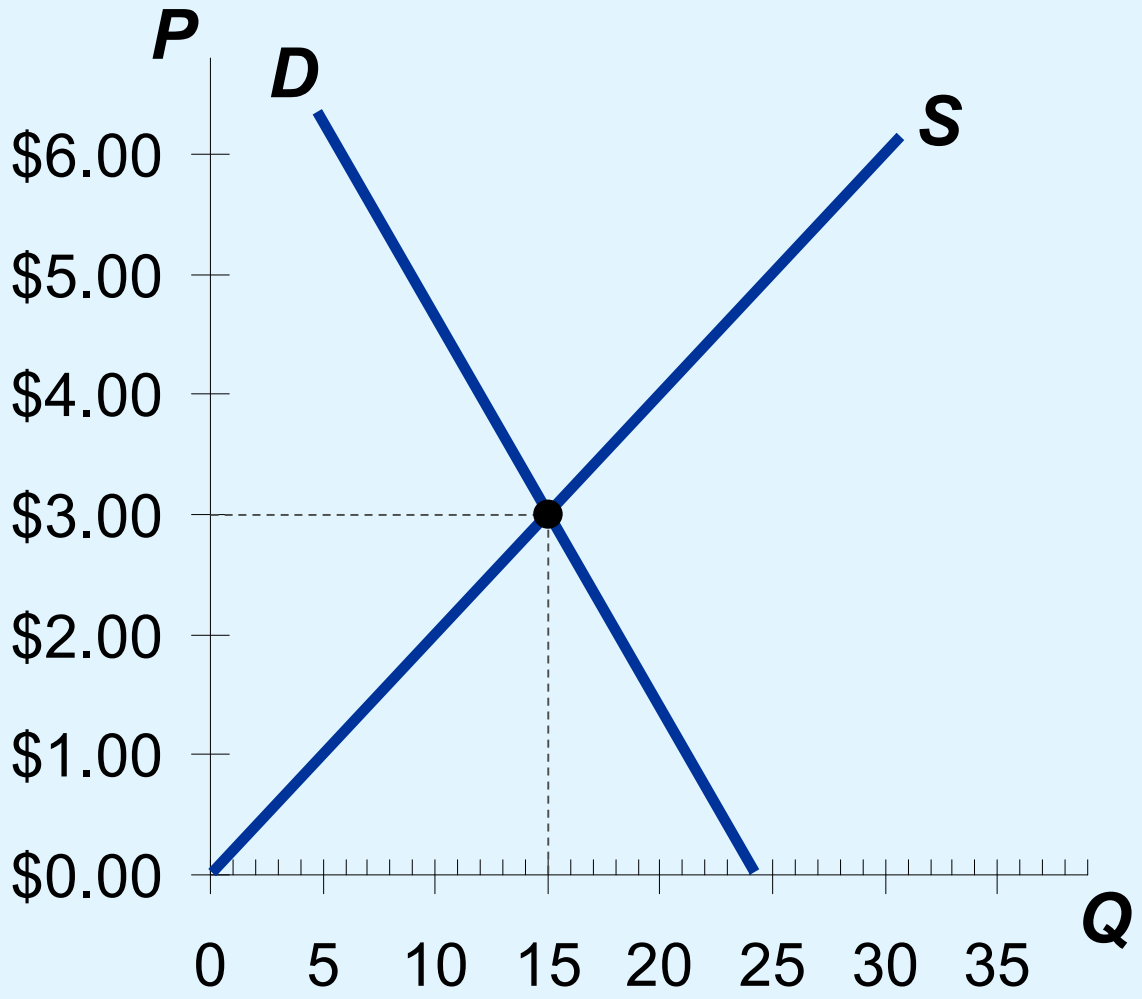


Trick question:

This shifts the demand curve for tax preparation software, not the supply curve.

Supply and Demand Together

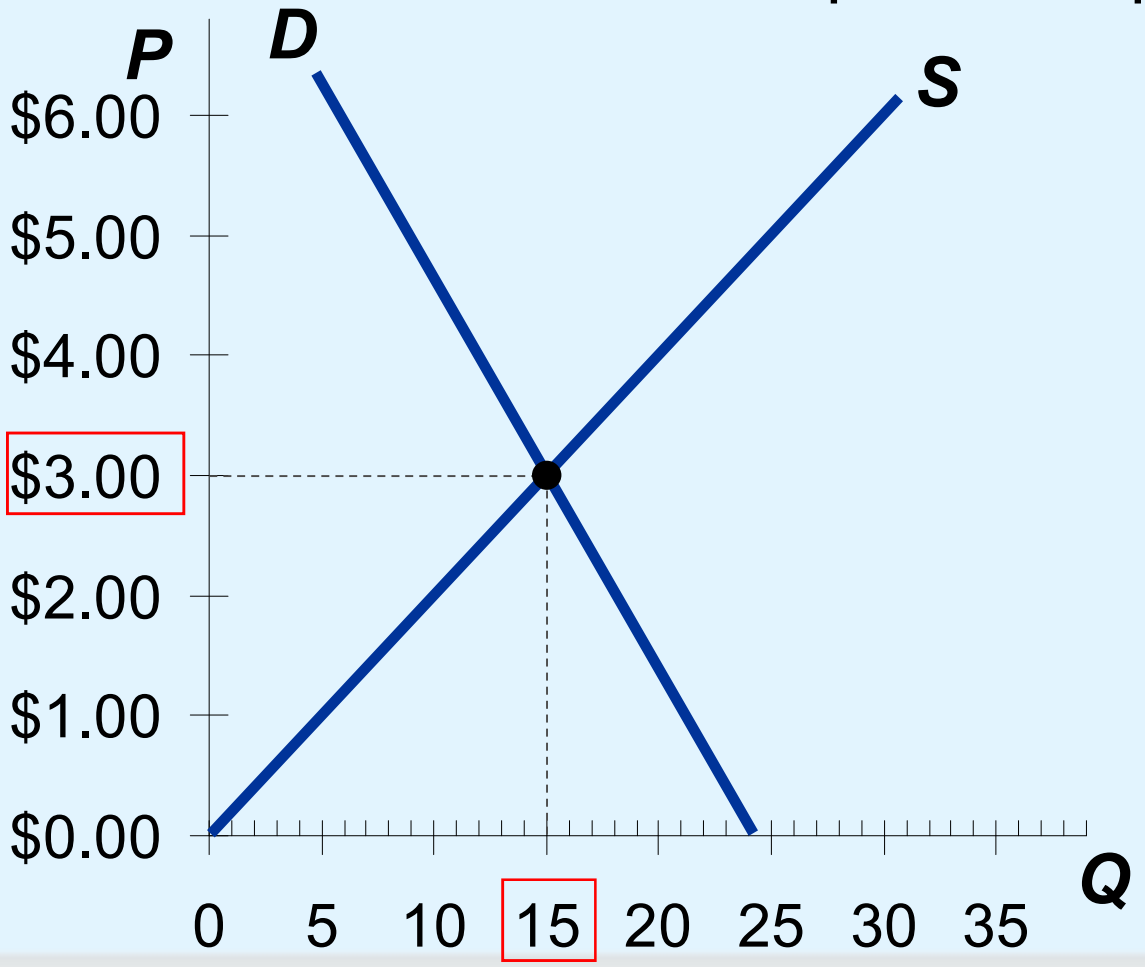
Equilibrium:
Price has reached the level where quantity supplied equals quantity demanded



Supply and Demand Together

Equilibrium price: price where Q supplied = Q demanded

Equilibrium quantity: Q supplied and demanded at the equilibrium price

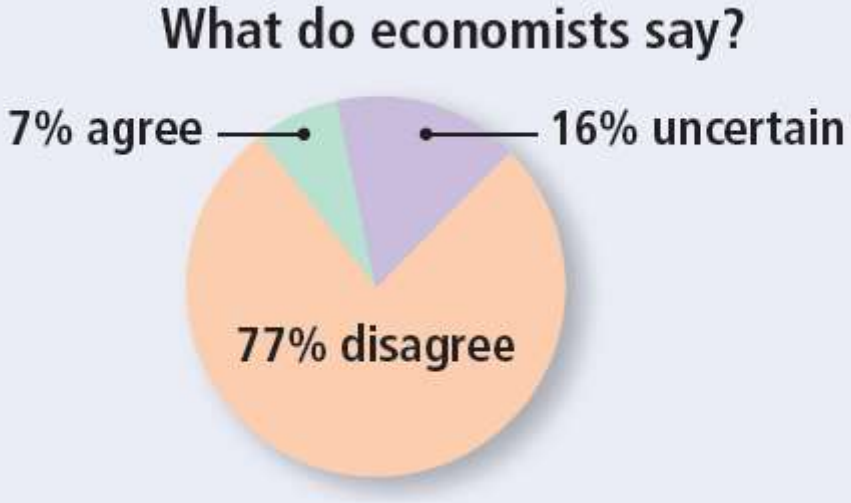


P	Q^D	Q^S
$0	24	0
1	21	5
2	18	10
3	15	15
4	12	20
5	9	25
6	6	30

ASK THE EXPERTS

Price Gouging

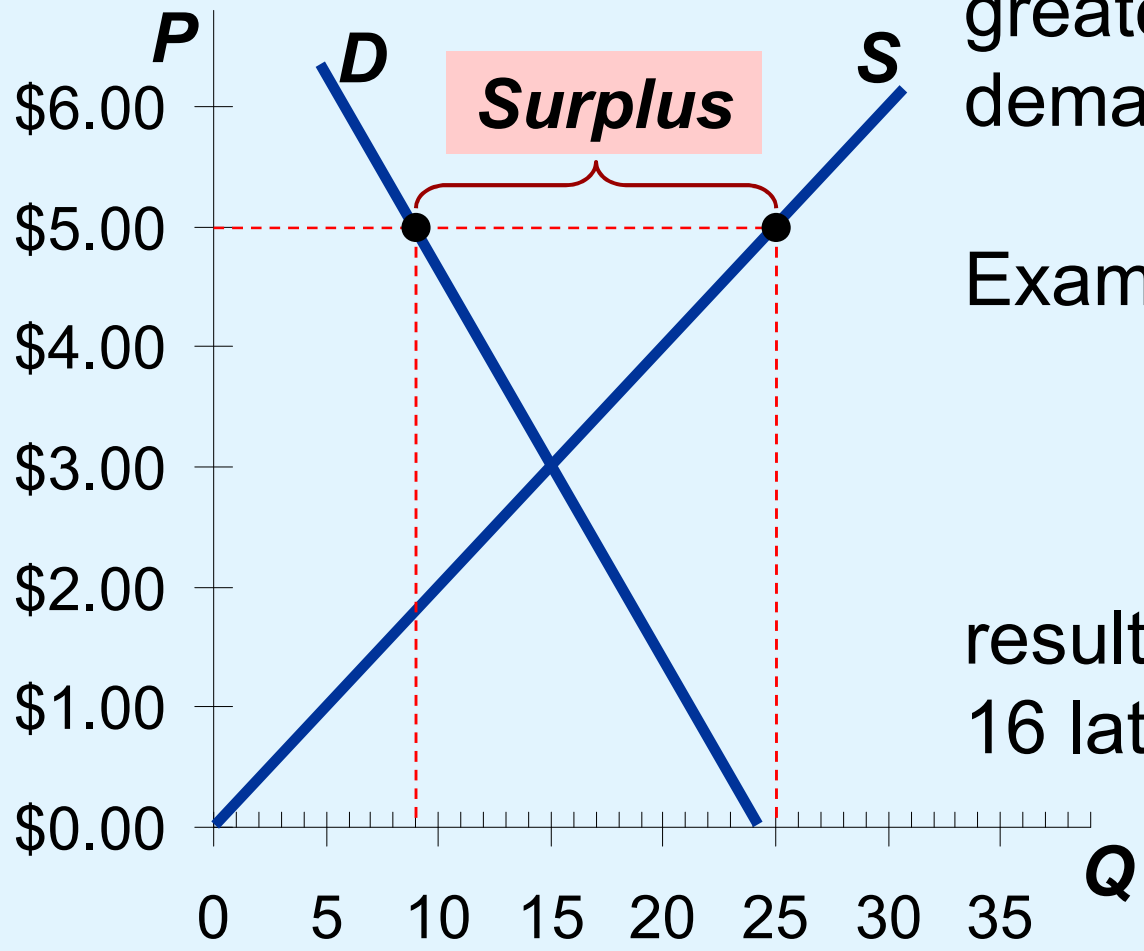
"Connecticut should pass its Senate Bill 60, which states that during a 'severe weather event emergency, no person within the chain of distribution of consumer goods and services shall sell or offer to sell consumer goods or services for a price that is unconscionably excessive.'"



Question

- “An increase in the demand for notebooks raises the quantity of notebooks demanded but not the quantity supplied.”
- Is this statement true or false?

Markets Not in Equilibrium: Surplus

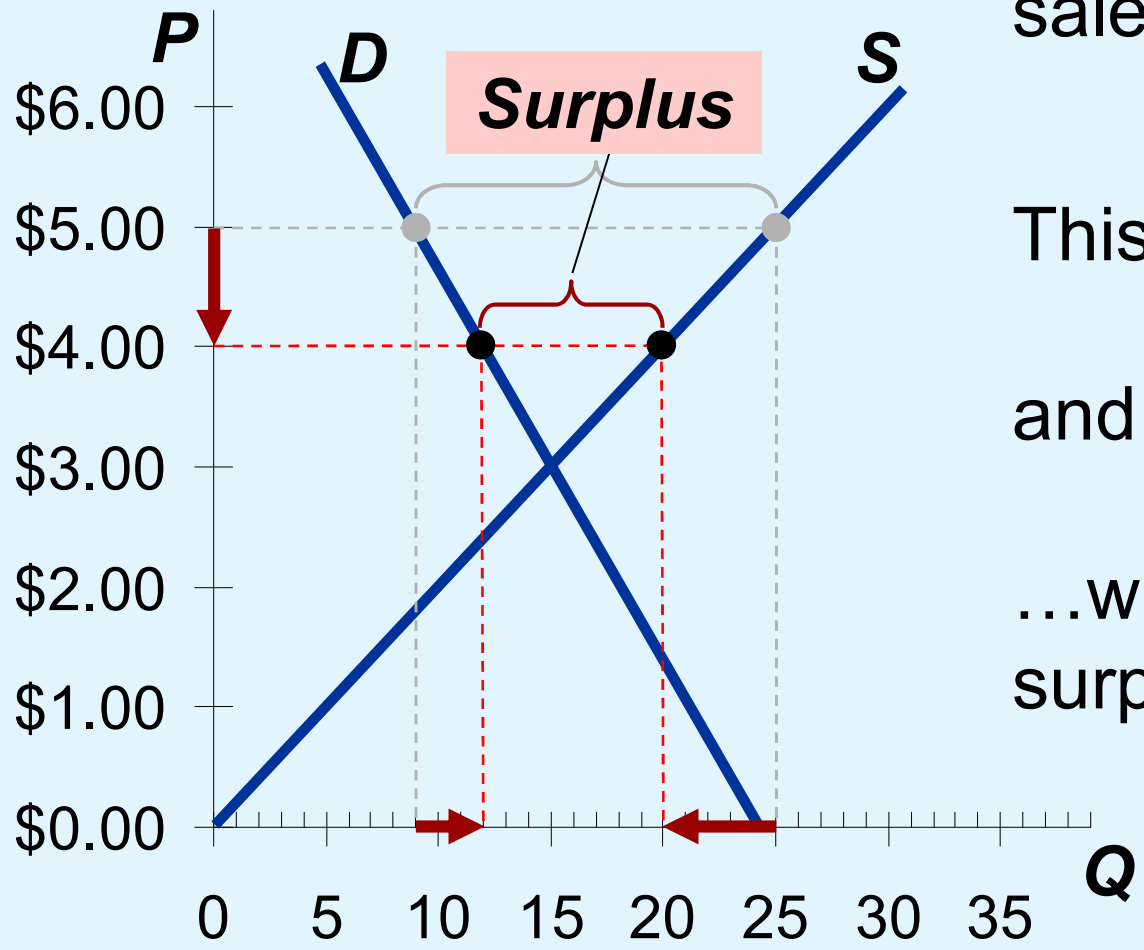


Surplus (excess supply): quantity supplied is greater than quantity demanded

Example: if P = \$5,
then Q^D = 9 lattes
and Q^S = 25 lattes

resulting in a surplus of 16 lattes

Markets Not in Equilibrium: Surplus



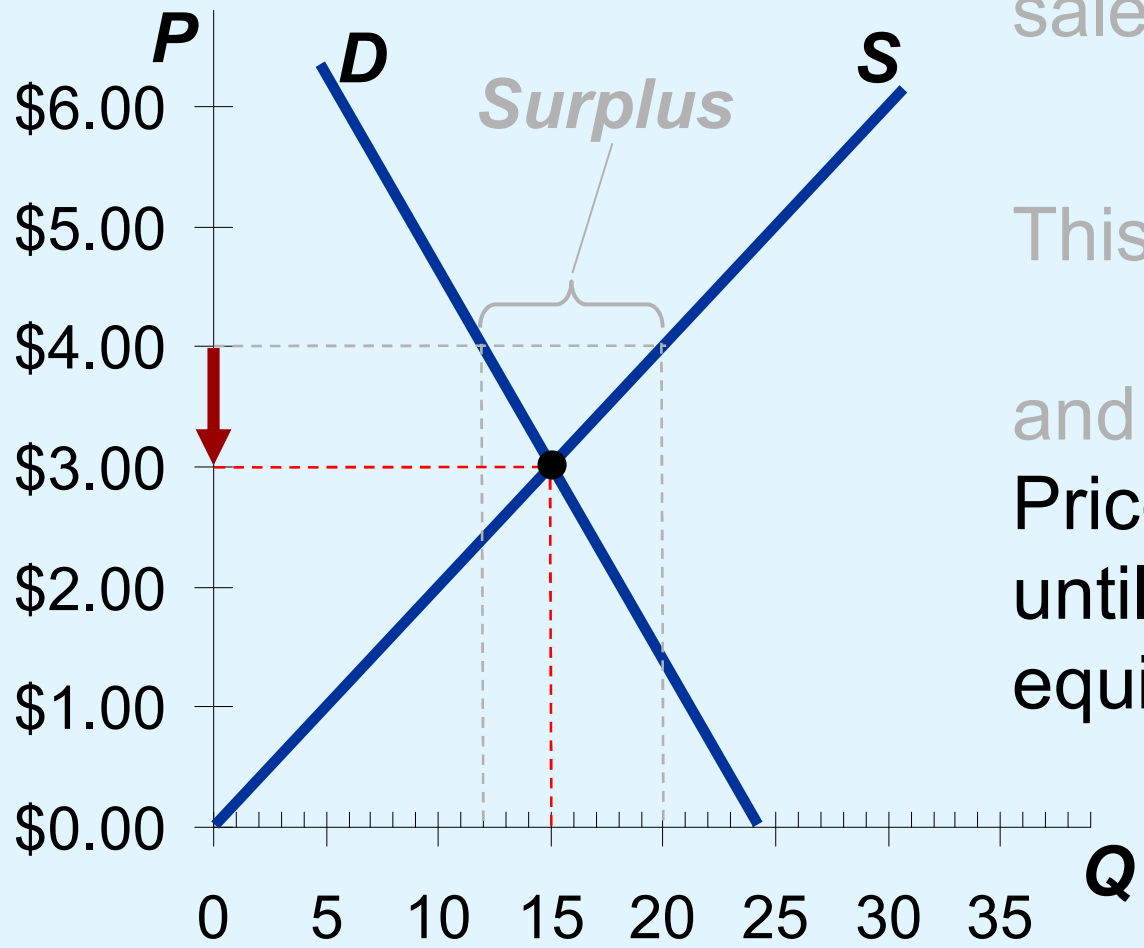
Facing a surplus, sellers try to increase sales by cutting price.

This causes Q^D to rise

and Q^S to fall…

…which reduces the surplus.

Markets Not in Equilibrium: Surplus



Facing a surplus, sellers try to increase sales by cutting price.

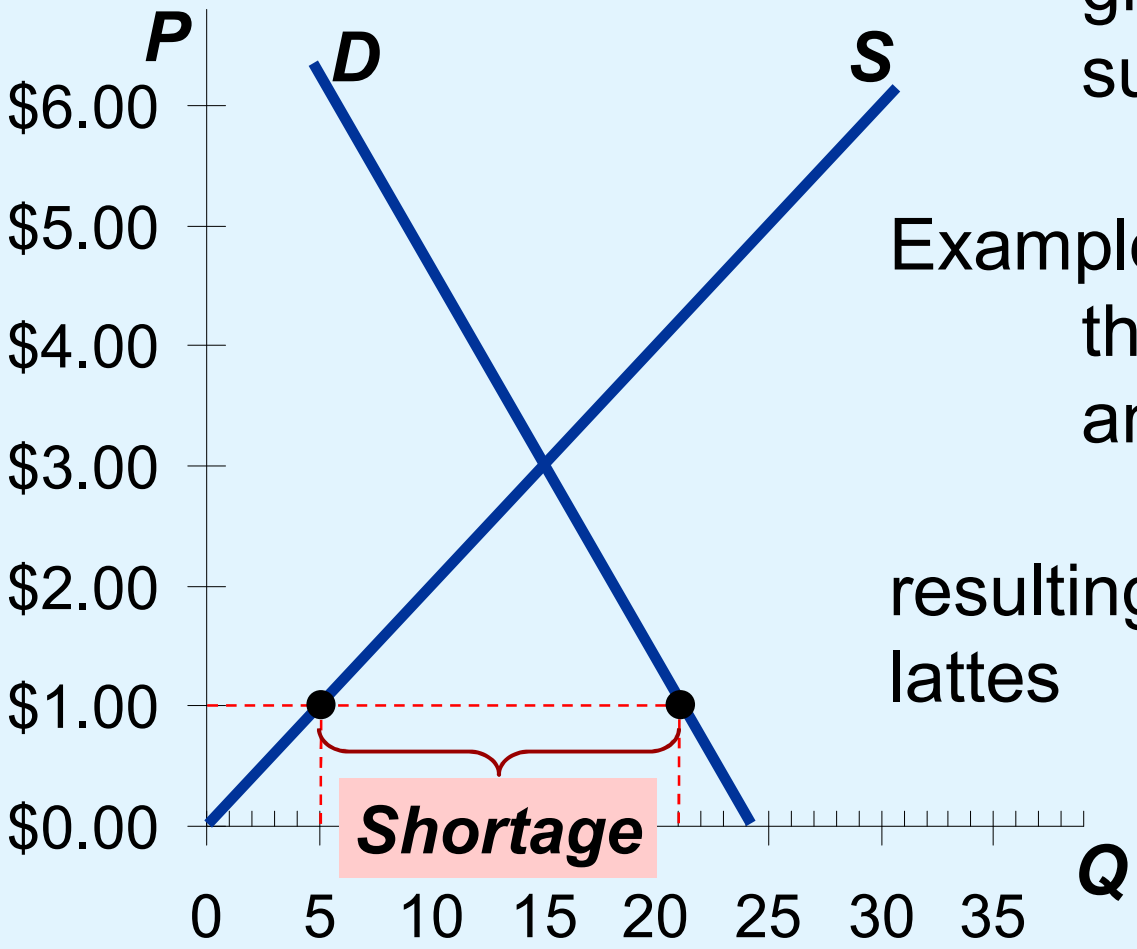
This causes Q^D to rise

and Q^S to fall…

Prices continue to fall until market reaches equilibrium.

Markets Not in Equilibrium: Shortage

Shortage (excess demand): quantity demanded is greater than quantity supplied



Example: if P = \$1, then Q^D = 21 lattes and Q^S = 5 lattes

resulting in a shortage of 16 lattes

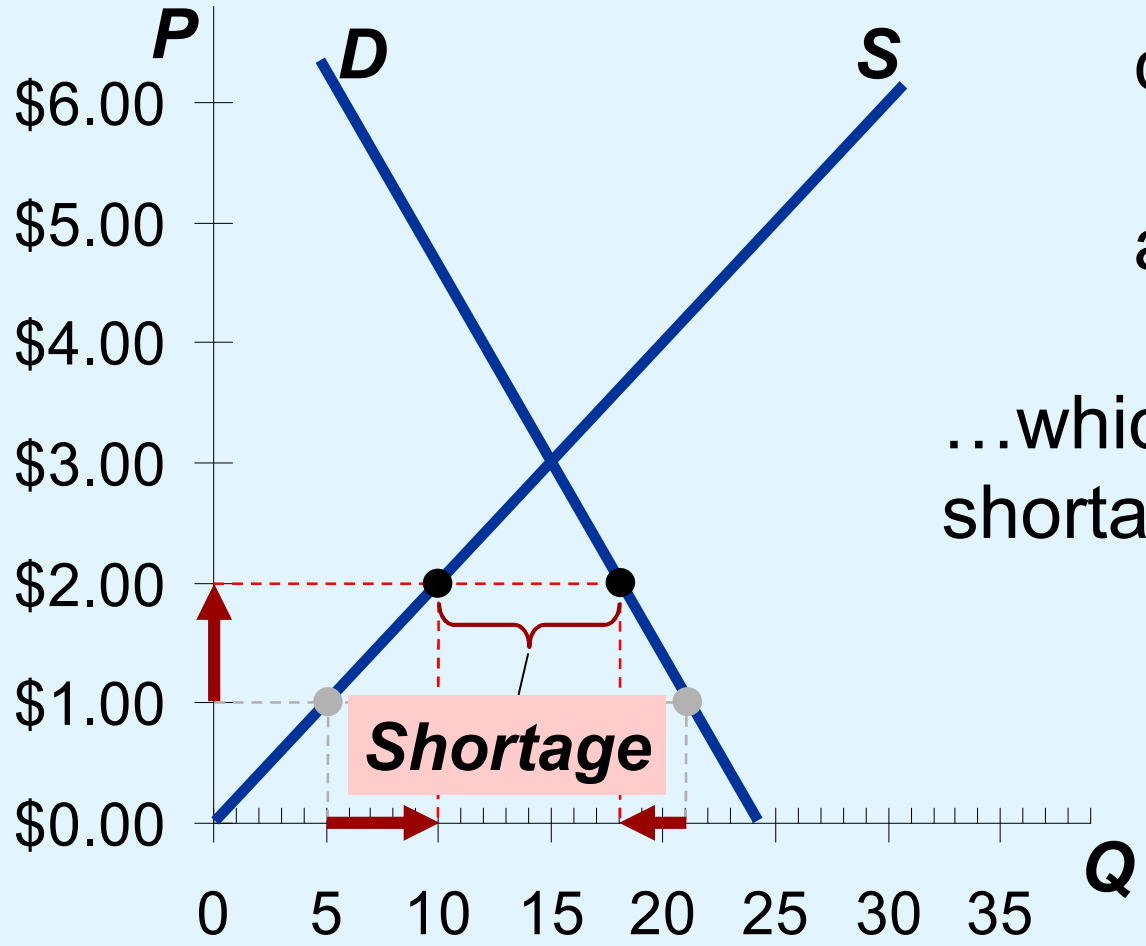
Markets Not in Equilibrium: Shortage

Facing a shortage, sellers raise the price,

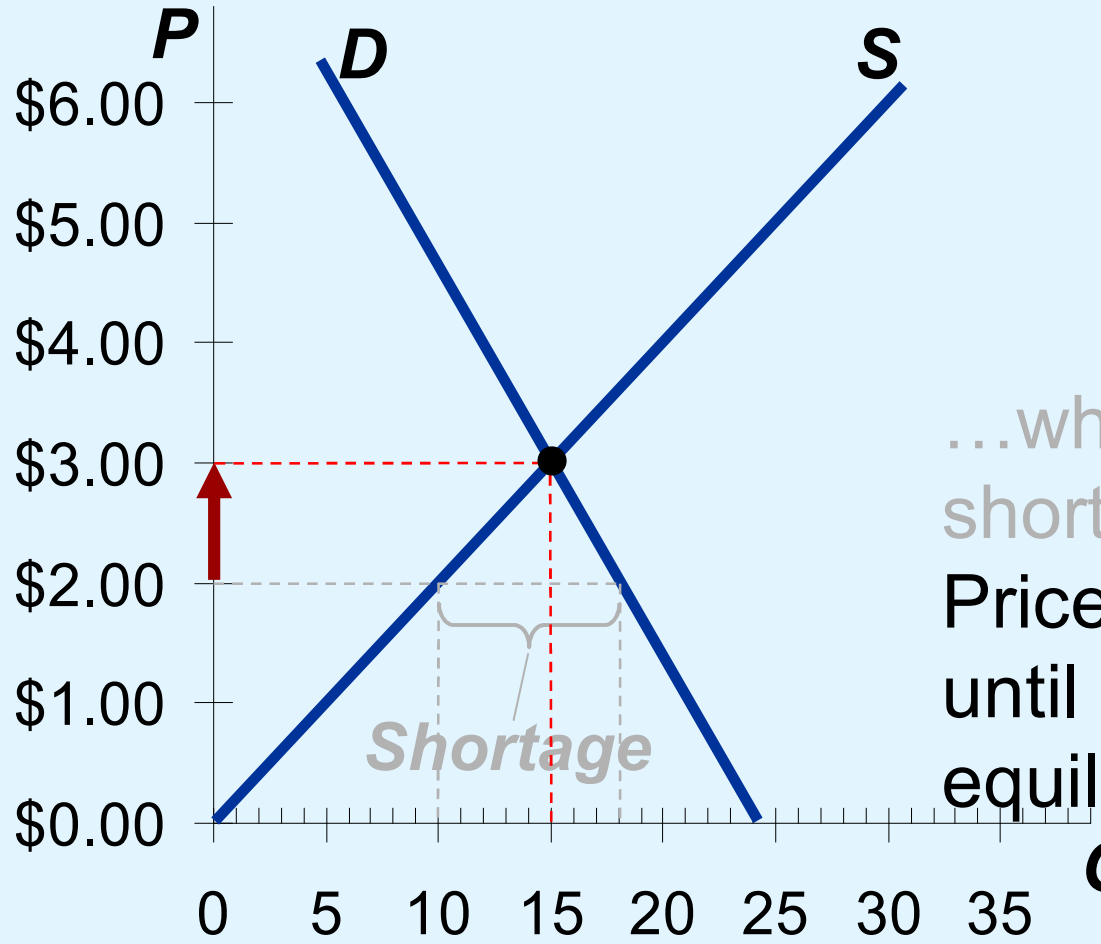
causing Q^D to fall

and Q^S to rise,

…which reduces the shortage.



Markets Not in Equilibrium: Shortage



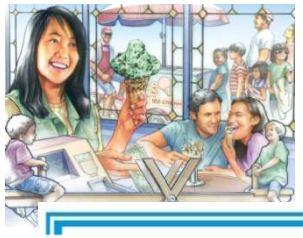
Facing a shortage, sellers raise the price,

causing Q^D to fall

and Q^S to rise,

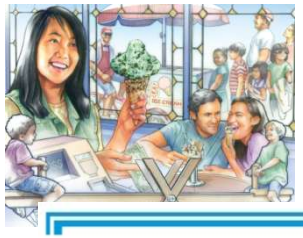
…which reduces the shortage.

Prices continue to rise until market reaches equilibrium.



Question

- Consider a competitive market for pizza in which the equilibrium price is 6 US dollars
- If the actual price in this market were above the equilibrium price, what would drive the market toward the equilibrium ?
- If the actual price in this market were below the equilibrium price, what would drive the market toward the equilibrium?

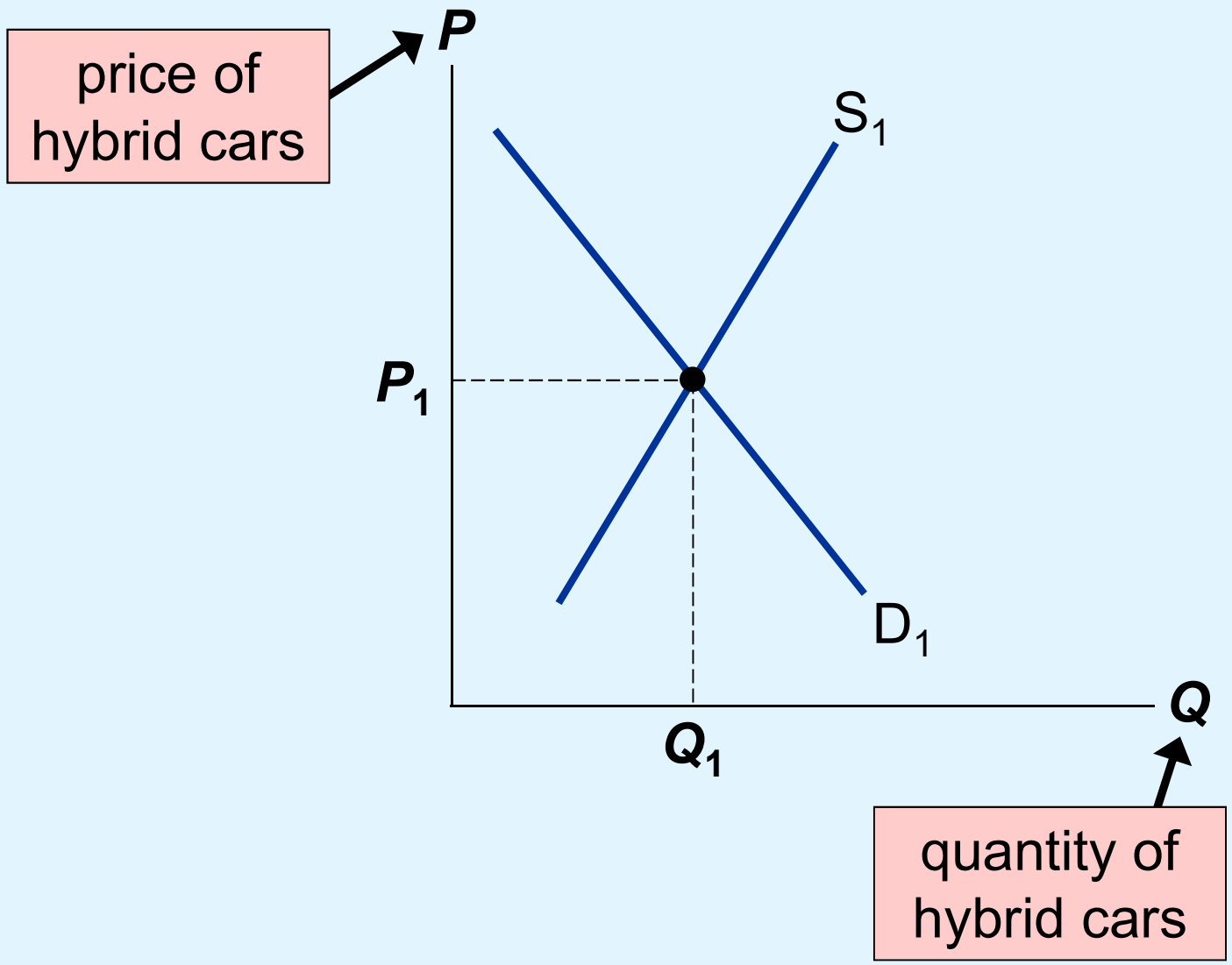


Supply and Demand Together

Three steps to analyzing changes in equilibrium

1. Decide whether the event shifts the supply curve, the demand curve, or, in some cases, both curves
2. Decide whether the curve shifts to the right or to the left
3. Use the supply-and-demand diagram
   - Compare the initial and the new equilibrium
   - Effects on equilibrium price and quantity

EXAMPLE: The Market for Hybrid Cars



EXAMPLE 1: A Shift in Demand

Hybrid Card

EVENT TO BE ANALYZED:
Increase in the price of gas.

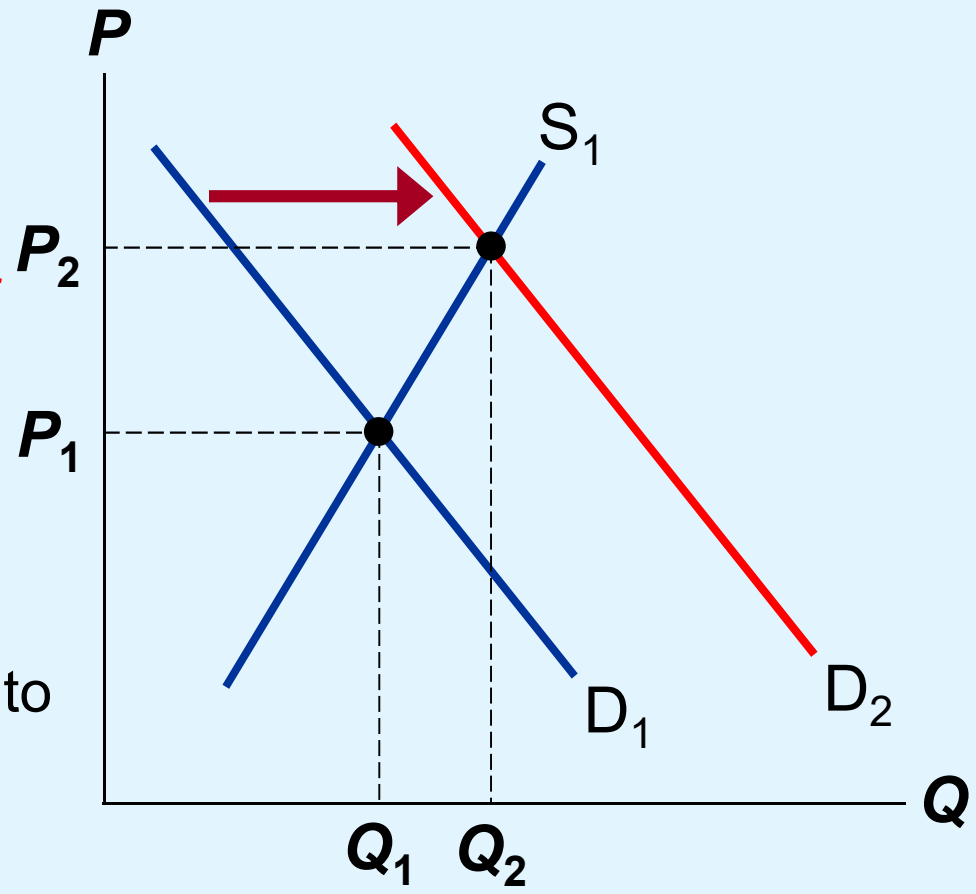
STEP 1: ***D*** curve shifts

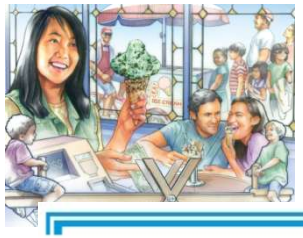
because price of gas affects demand for hybrids. (***S*** curve does not shift, because price of gas does not affect cost of producing hybrids)

STEP 2: ***D*** shifts right

because high gas price makes hybrids more attractive relative to other cars.

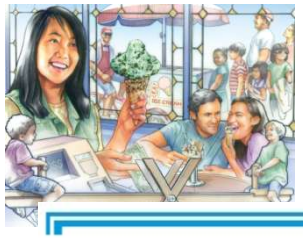
STEP 3: The shift causes an increase in price and quantity of hybrid cars.





Shift vs. Movement Along Curve

- Change in supply:
  - A shift in the S curve
  - Occurs when a non-price determinant of supply changes (like technology or costs)
- Change in the quantity supplied:
  - A movement along a fixed S curve
  - Occurs when P changes



Shift vs. Movement Along Curve

- Change in demand:
  - A shift in the D curve
  - Occurs when a non-price determinant of demand changes (like income or # of buyers)
- Change in the quantity demanded:
  - A movement along a fixed D curve
  - Occurs when P changes

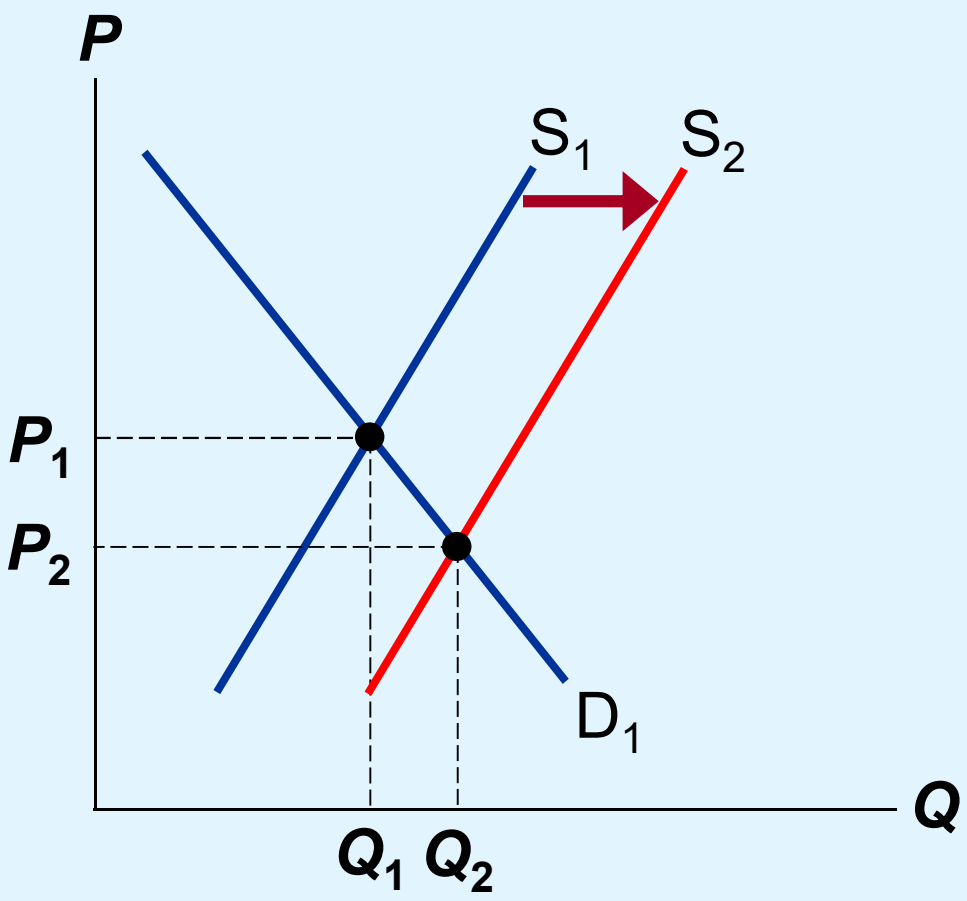
EXAMPLE 2: A Shift in Supply

EVENT: New technology reduces cost of producing hybrid cars.

STEP 1: ***S*** curve shifts because event affects cost of production. (***D*** curve does not shift, because production technology is not one of the factors that affect demand)

STEP 2: ***S*** shifts <u>right</u> because event reduces cost, makes production more profitable at any given price.

STEP 3: The shift causes price to fall and quantity to rise.



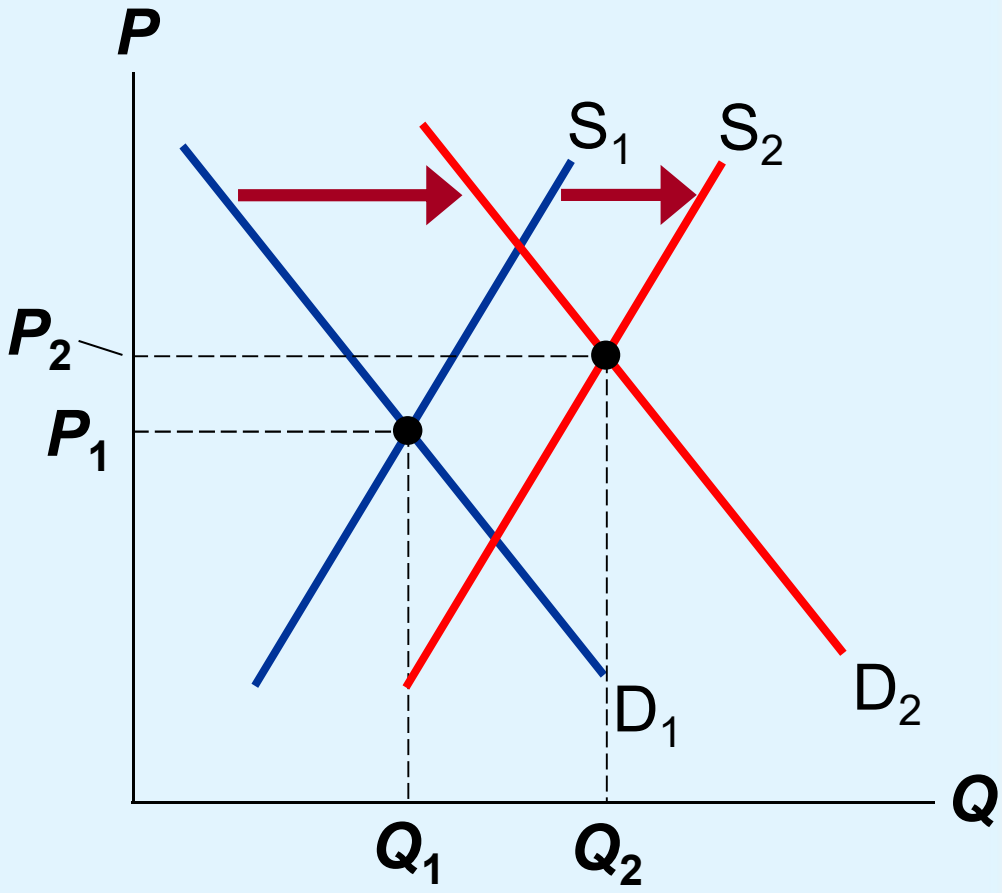
EXAMPLE 3: A Shift in Both Supply and Demand

EVENTS: Price of gas rises AND new technology reduces production costs

STEP 1: Both curves shift.
STEP 2: Both shift to the right.

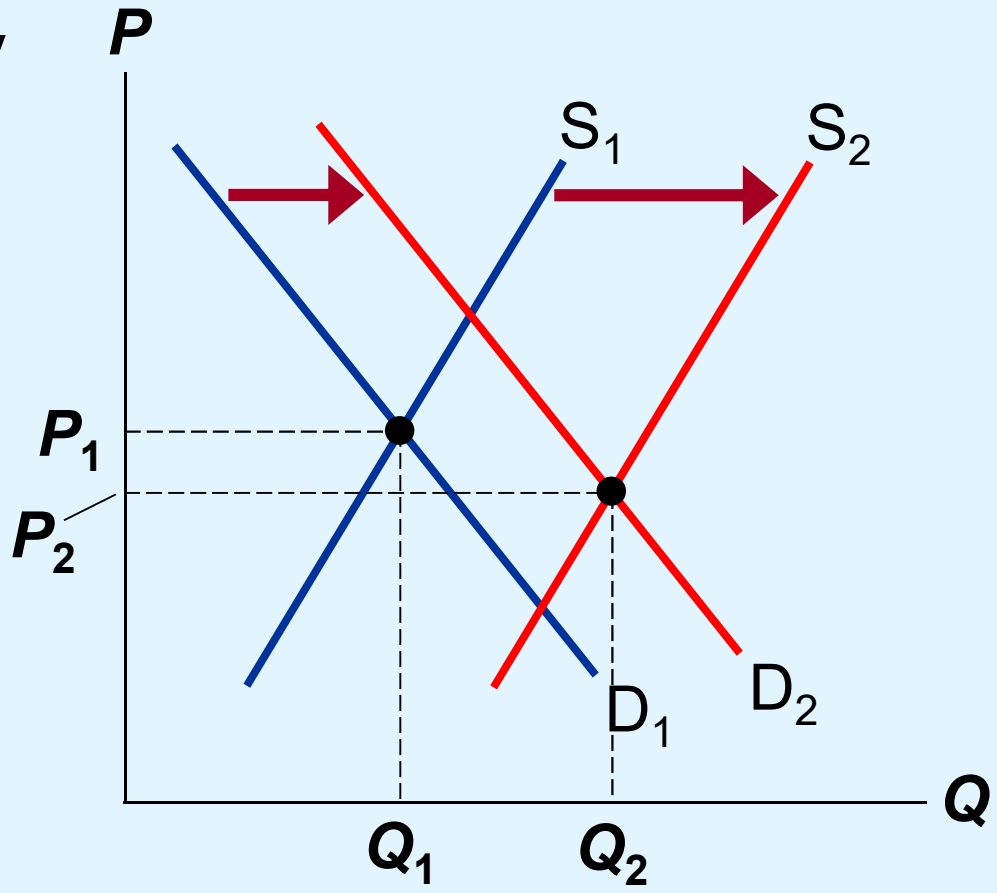
STEP 3: ***Q*** rises, but the effect on ***P*** is ambiguous:

If demand increases more than supply, ***P*** rises.



EXAMPLE 3: A Shift in Both Supply and Demand

EVENTS: Price of gas rises AND new technology reduces production costs



STEP 3: ***Q*** rises, but the effect on ***P*** is ambiguous:

But if supply increases more than demand, ***P*** falls.

Active Learning 3 Shifts in supply and demand

Use the three-step method to analyze the effects of each event on the equilibrium price and quantity of music downloads.

Event A: A fall in the price of *music CDs*

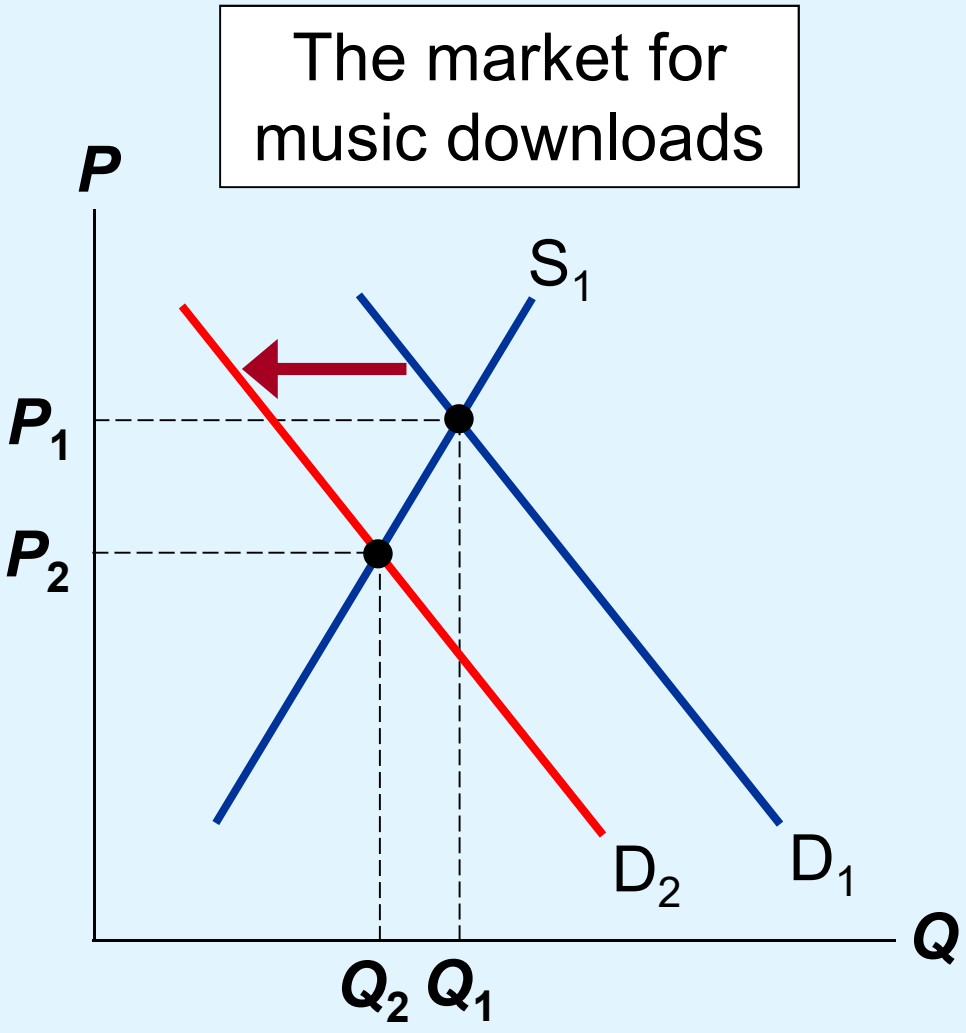
Event B: Sellers of music downloads negotiate a reduction in the royalties they must pay for each song they sell.

Event C: Events A and B both occur.

Active Learning 3 **A.** A fall in the price of music CDs

STEPS:

1. ***D*** curve shifts
2. ***D*** curve shifts left
3. ***P*** and ***Q*** both fall



Active Learning 3

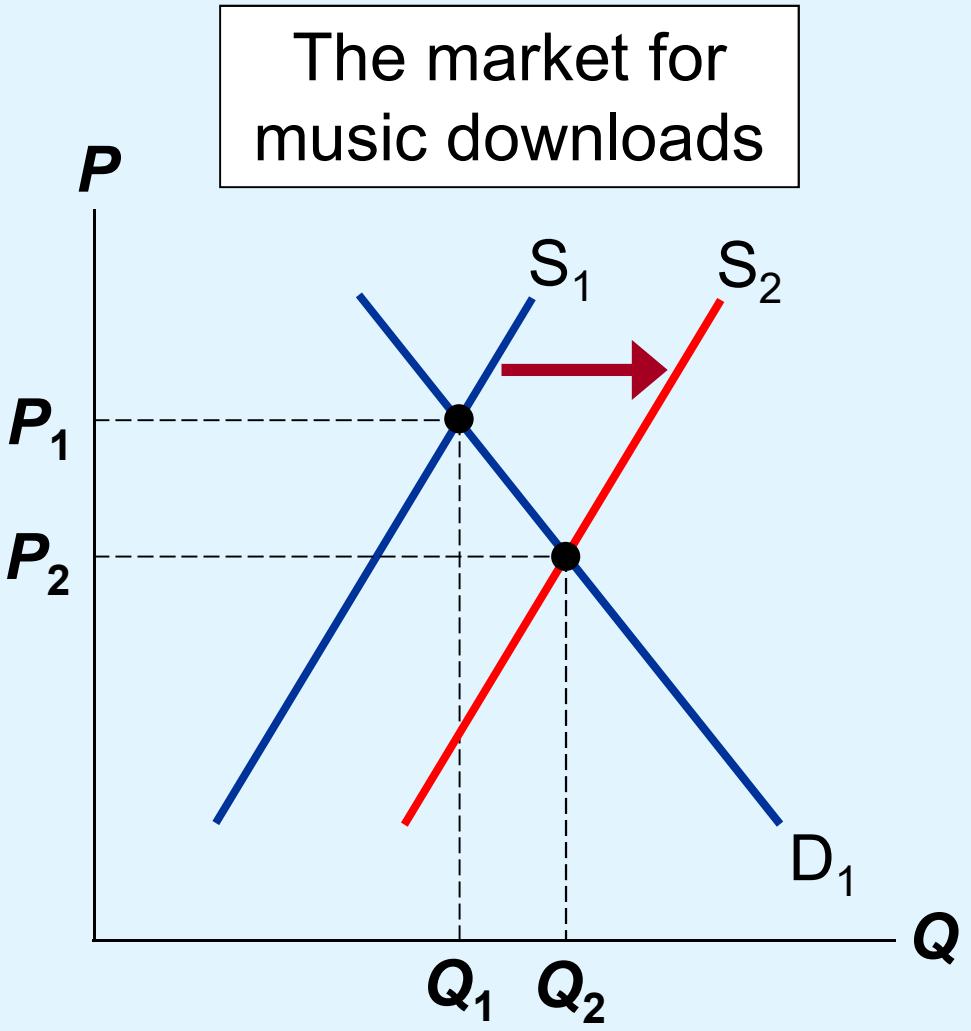
B. Fall in cost of royalties

STEPS:

1. ***S*** curve shifts (Royalties are part of sellers' costs)

2. ***S*** curve shifts right

3. ***P*** falls, ***Q*** rises



Active Learning 3

C. Fall in price of music CDs and fall in cost of royalties

STEPS:

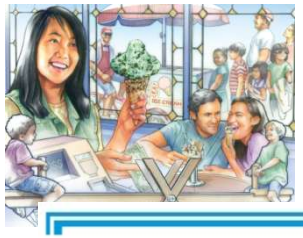
1. Both curves shift (see parts A & B)

2. D shifts left, S shifts right

3. P falls.

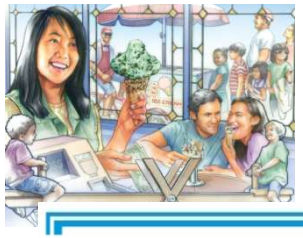
   Effect on Q is ambiguous:

   - the fall in demand reduces Q,
   - the increase in supply increases Q.



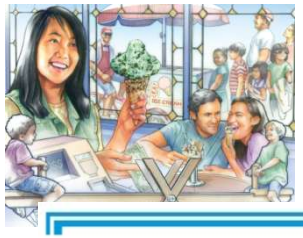
How Prices Allocate Resources

- "Markets are usually a good way to organize economic activity"
- In market economies
  - Prices adjust to balance supply and demand
- These equilibrium prices
  - Are the signals that guide economic decisions and thereby allocate scarce resources



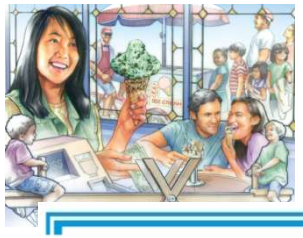
Question

- Consider the market for minivans
- For each of the events listed here, identify which of the determinants of demand or supply are affected
- Also indicate whether demand or supply increases or decreases.
- Then draw a diagram to show the effect on the price and quantity of minivans
  - People decide to have more children
  - A strike by steelworkers raises steel prices
  - Engineers develop new automated machinery for the production of minivans
  - The price of sports utility vehicles (運動型休旅車) rises
  - A stock market crash lowers people' wealth



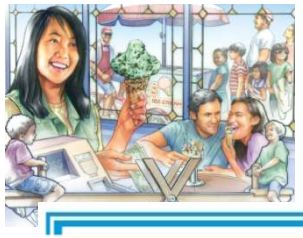
Question

- Over the past 40 years, technological advances have reduced the cost of computer chips.
- How do you think this has affected that market for computers? For computer software? For typewriters?



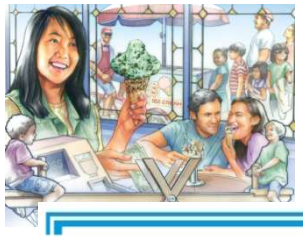
Question

- Consider the markets for film streaming services, TV screens, and tickets at movie theaters.
- For each pair, identify whether they are complements or substitutes
  - Film streaming and TV screens
  - Film streaming and movie tickets
  - TV screens and movie tickets
- How does the change in the market for TV screens affect the markets for film streaming and movie tickets?



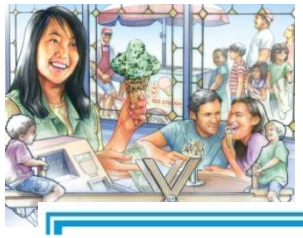
Question

- Ketchup is a complement (as well as a condiment) for hot dogs.
- If the price of hot dogs rises, what happens in the market for ketchup?
  - For tomatoes?
  - For tomato juice?
  - For orange juice?



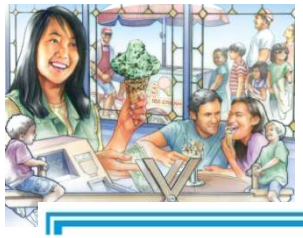
Question

- Consider the following events:
  - Scientists reveal that eating oranges decreases the risk of diabetes, and at the same time, farmers use a new fertilizer that makes orange trees produce more oranges.
- How about the equilibrium price and quantity of oranges?



Why did the Price of Coffee Rise in 2014?





Eye on the Price of Coffee (1 of 3)

Why Did the Price of Coffee Rise in 2014?

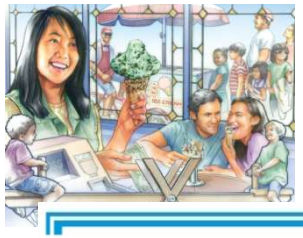
When a fungus called coffee rust swept though Brazil and other countries of South America in 2014, world coffee production decreased and the price of coffee beans increased.

The table provides some data on the quantity and price of coffee in 2013 and 2014.

The table tell us that the quantity of coffee decreased in 2014 and the price rose.

	Quantity of Coffee	Price of Coffee
Year	(billions of pounds per year)	(dollars per pound)
2013	19.4	1.04
2014	18.7	1.35

咖啡葉銹病
真菌感染



Eye on the Price of Coffee (2 of 3)

Why Did the Price of Coffee Rise in 2014?

Did the price rise because demand increased?

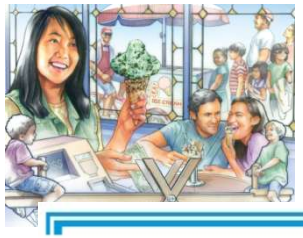
Or did the price rise because supply decreased?

You know that when demand increases, the price rises but the quantity bought increases.

You also know that when supply decreases, the price rises and the quantity bought decreases.

So: Did the demand for coffee increase? Or did the supply of coffee decrease?

Because the quantity bought decreased when the price rose in 2014, the supply of coffee decreased in 2014.



Eye on the Price of Coffee (3 of 3)

Why Did the Price of Coffee Rise in 2014?

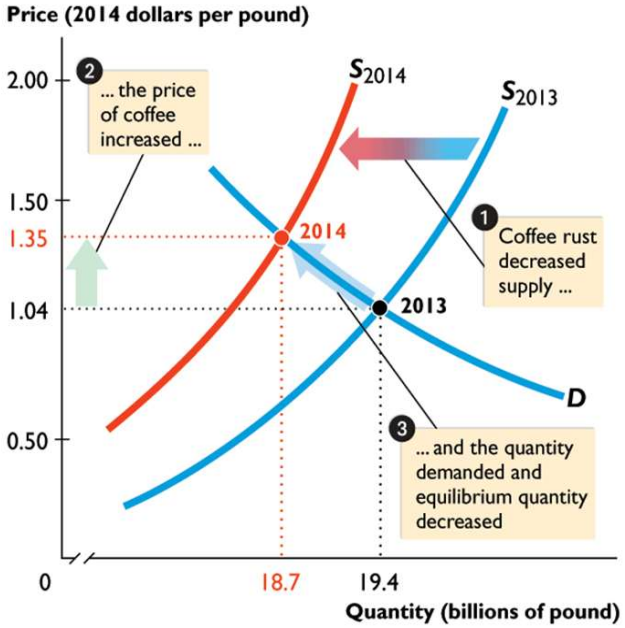
The figure shows the market for coffee in 2013.

The price was $1.04 a pound and 19.4 billion pounds were bought.

In 2014, supply decreased.

The price rose to $1.35 a pound and 18.7 billion pounds were bought.

The quantity demanded decreased along the demand curve.



The Market for Coffee in 2013–2014

Summary

- Economists use the model of supply and demand to analyze competitive markets.
  - Many buyers and sellers, all are price takers
- The demand curve shows how the quantity of a good demanded depends on the price.
  - Law of demand: as the price of a good falls, the quantity demanded rises; the ***D*** curve slopes downward
  - Other determinants of demand: income, prices of substitutes and complements, tastes, expectations, and number of buyers.
  - If one of these factors changes, the ***D*** curve shifts

Summary

- The supply curve shows how the quantity of a good supplied depends on the price.
  - Law of supply: as the price of a good rises, the quantity supplied rises; the ***S*** curve slopes upward.
- Other determinants of supply: input prices, technology, expectations, and number of sellers.
  - If one of these factors changes, supply curve shifts.
- The intersection of the supply and demand curves determines the market equilibrium.
  - At the equilibrium price, quantity demanded = quantity supplied

Summary

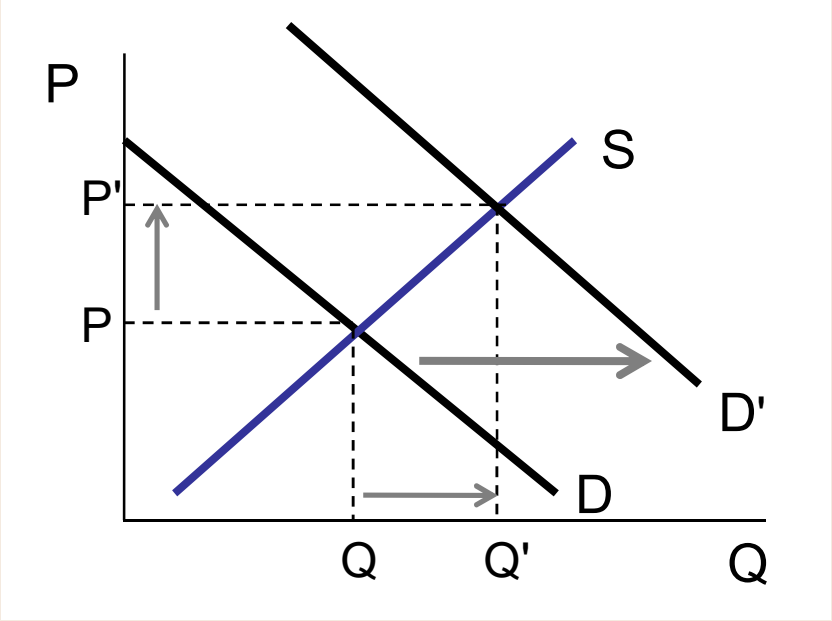
- The behavior of buyers and sellers naturally drives markets toward their equilibrium.
  - When the market price is above the equilibrium price, there is a surplus of the good, which causes the market price to fall.
  - When the market price is below the equilibrium price, there is a shortage, which causes the market price to rise.

Summary

- To analyze how any event influences a market, we use the supply-and-demand diagram to examine how the event affects the equilibrium price and quantity.
  1. Decide whether the event shifts the supply curve or the demand curve (or both).
  2. Decide in which direction the curve shifts.
  3. Compare the new equilibrium with the initial one.
- In market economies, prices are the signals that guide economic decisions and thereby allocate scarce resources.

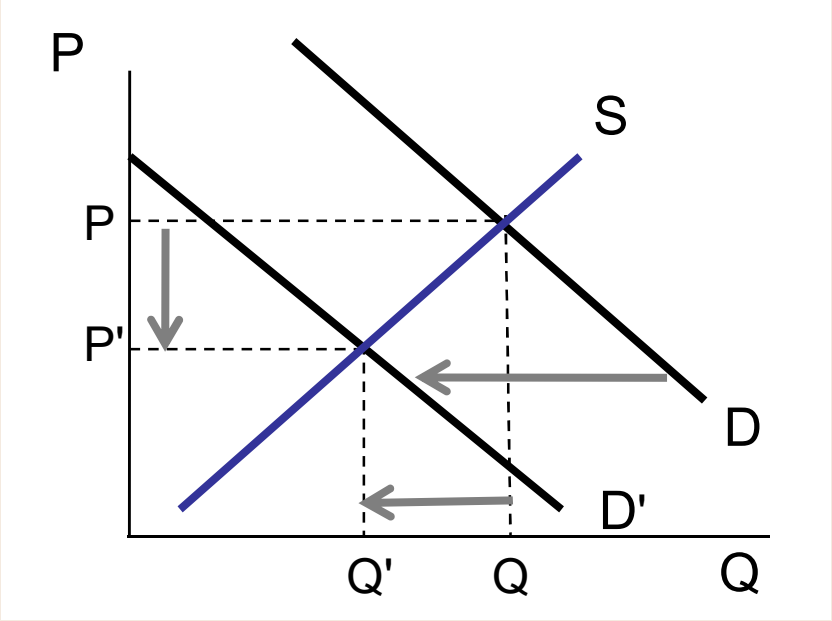
Supply and Demand Shifts: Four Rules

1. An increase in demand will lead to an increase in both equilibrium price and quantity



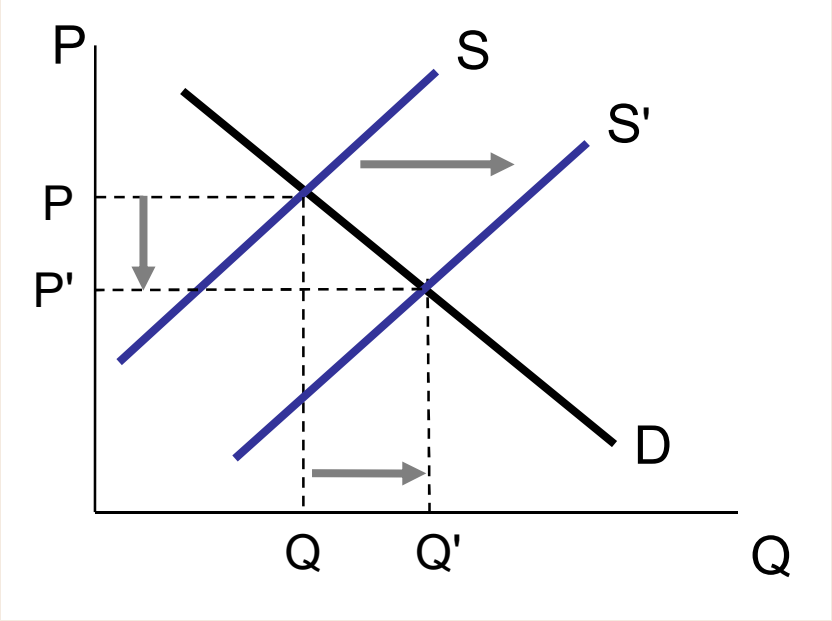
Supply and Demand Shifts: Four Rules

2. An decrease in demand will lead to a decrease in both equilibrium price and quantity



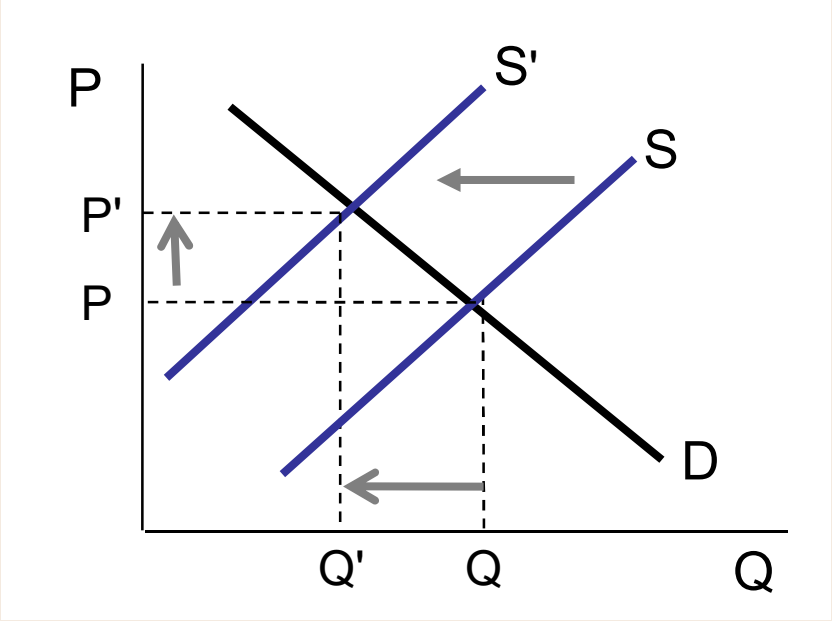
Supply and Demand Shifts: Four Rules

3. An increase in supply will lead to a decrease in the equilibrium price and an increase in the equilibrium quantity.



Supply and Demand Shifts: Four Rules

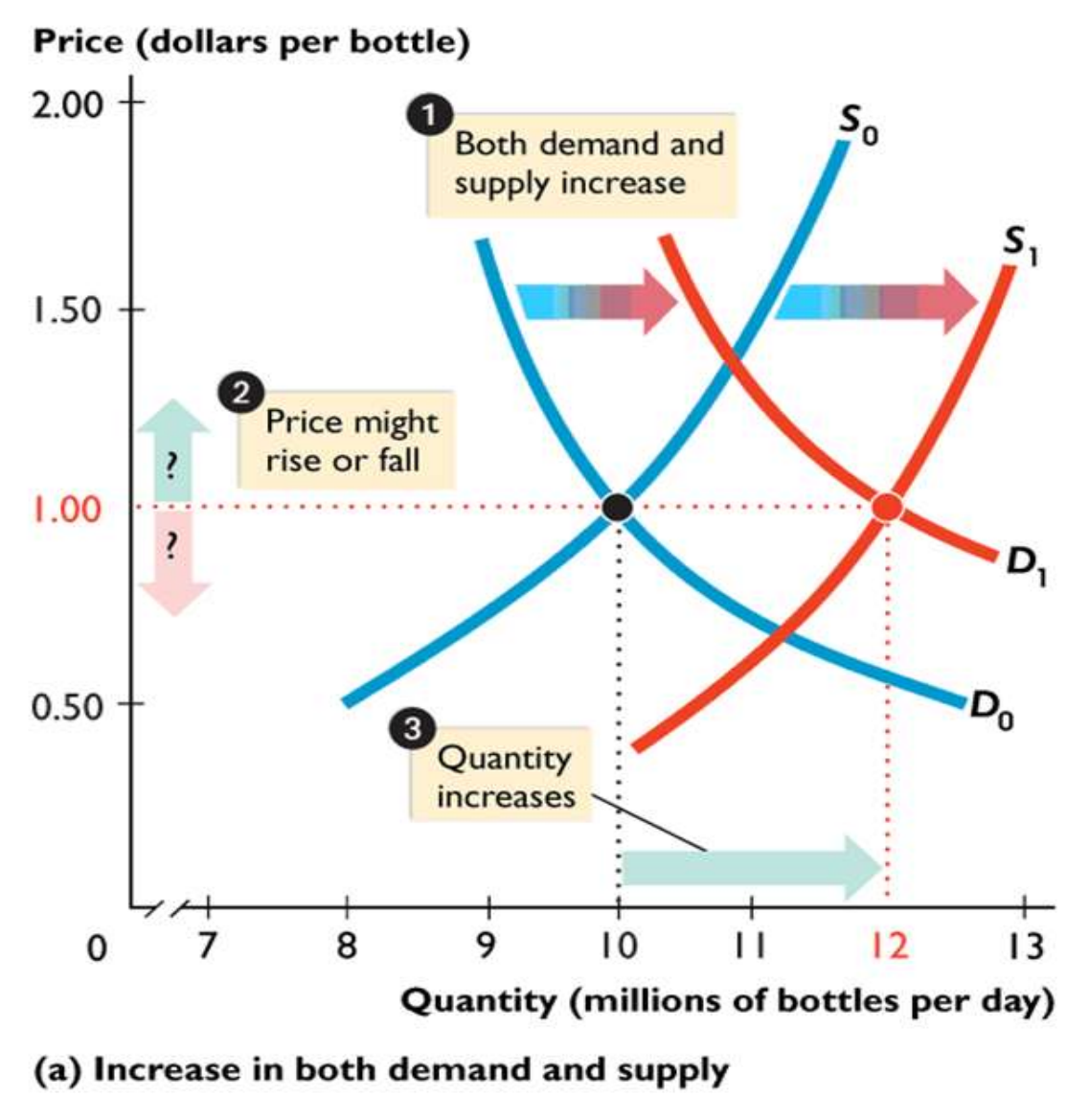
4. An decrease in supply will lead to an increase in the equilibrium price and a decrease in the equilibrium quantity.



Changes in Supply and Demand

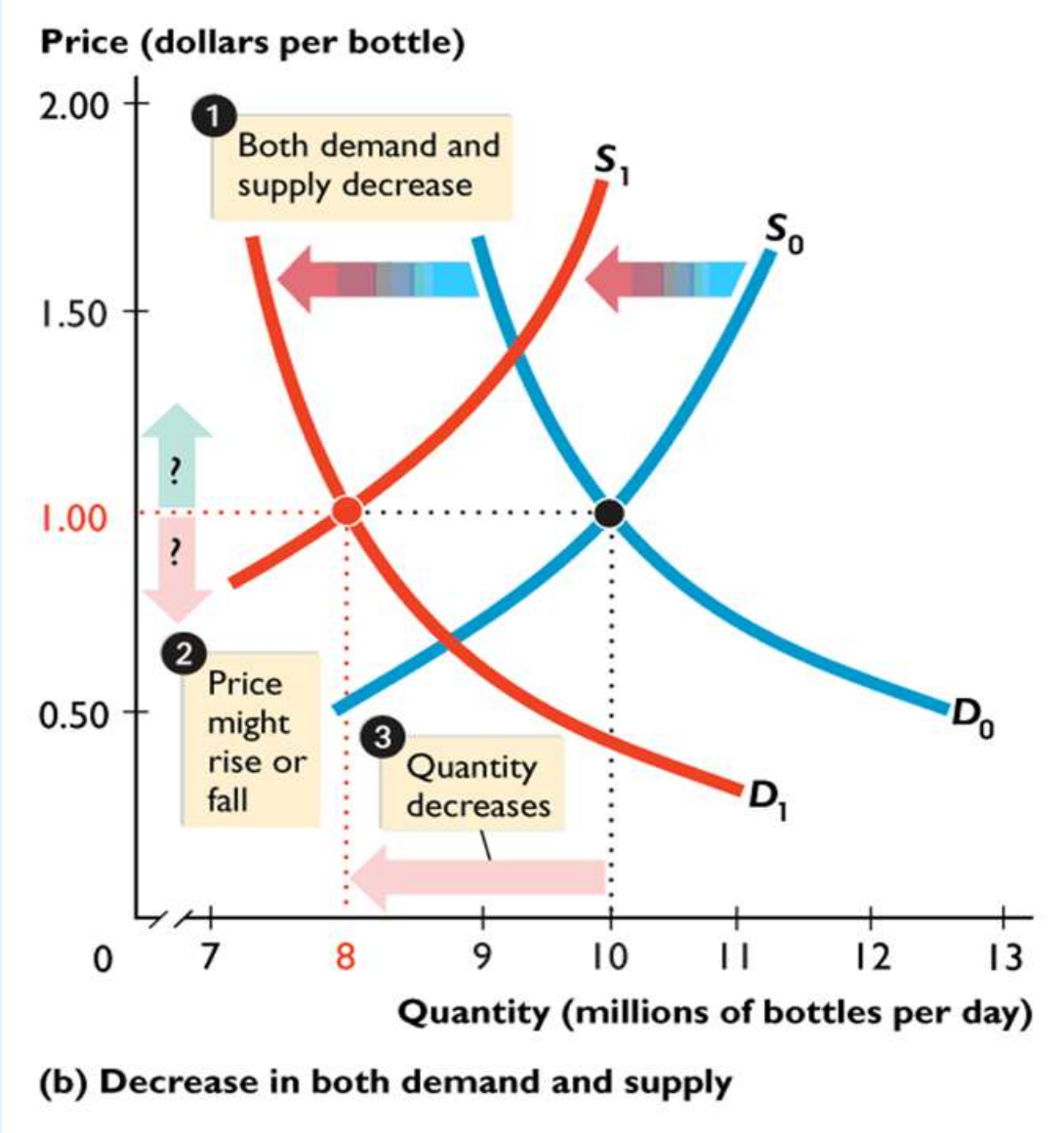
	Supply	
Demand	**Increases**	**Decreases**
Increases	P Depends Q Increases	P Increases Q Depends
Decreases	P Decreases Q Depends	P Depends Q Decreases

Increase in Both Demand and Supply



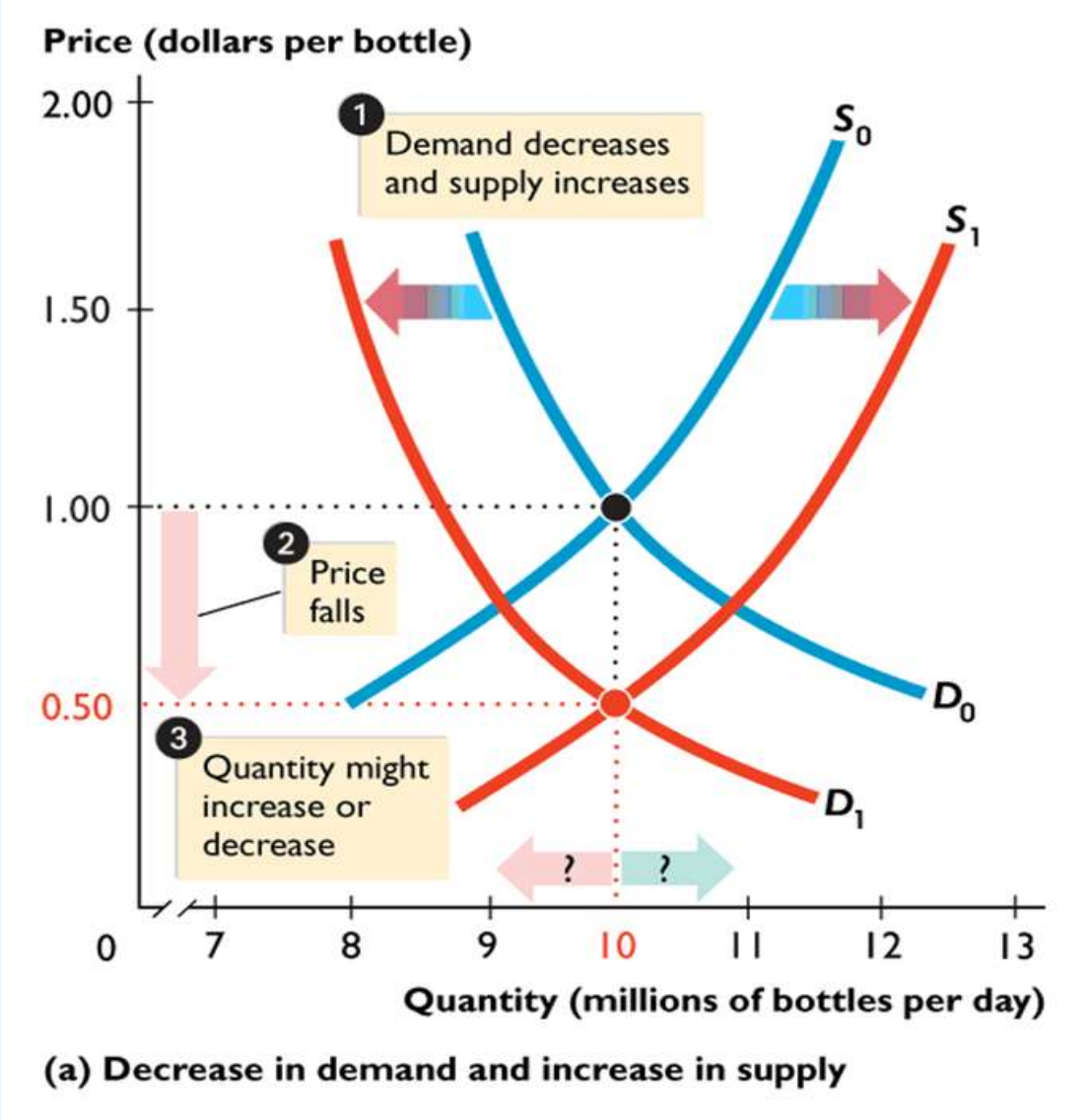
(a) Increase in both demand and supply

Decrease in Both Demand and Supply



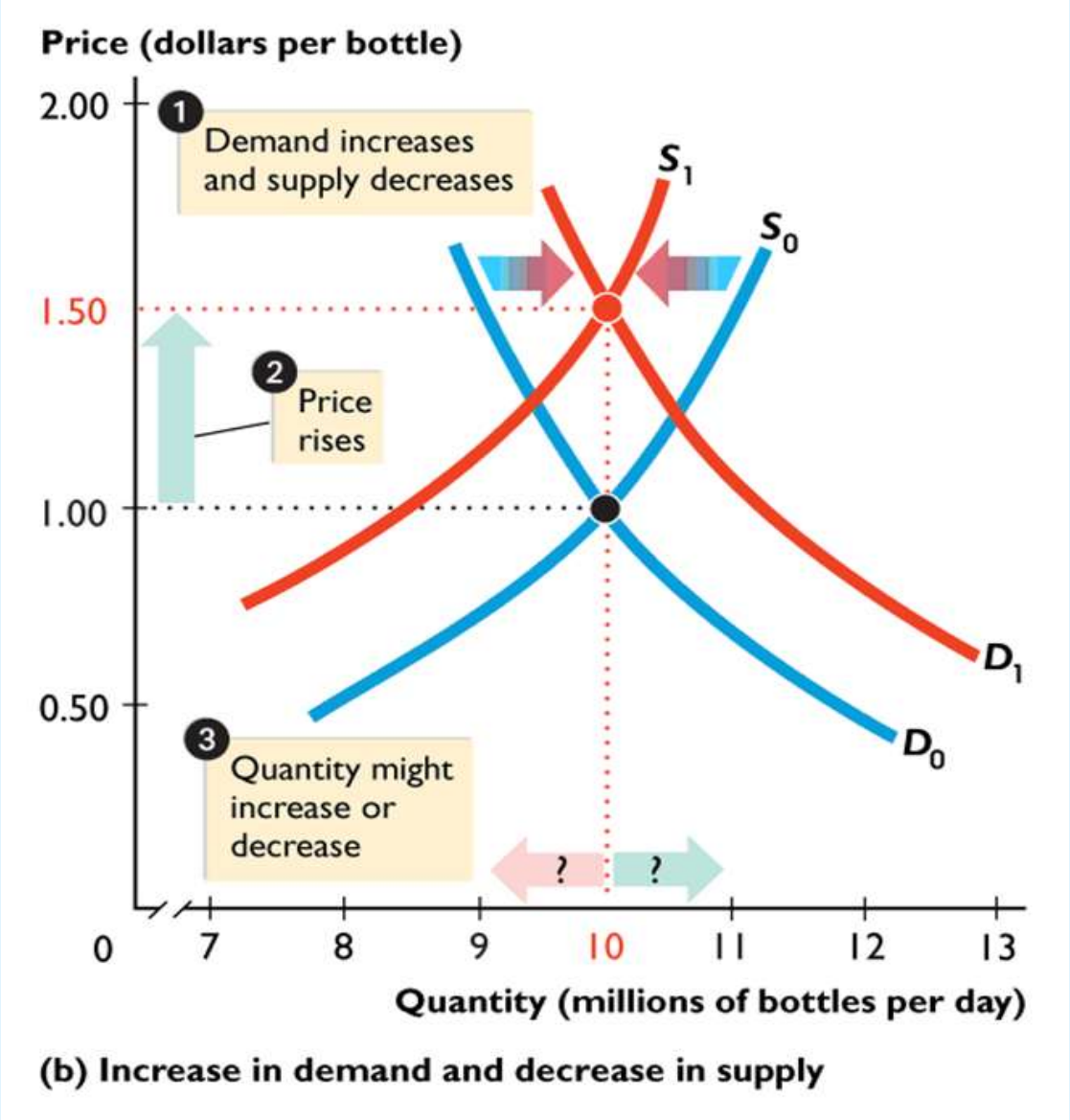
(b) Decrease in both demand and supply

Decrease in Demand but Increase in Supply



(a) Decrease in demand and increase in supply

Increase in Demand but Decrease in Supply



(b) Increase in demand and decrease in supply